New L J Institute of Engineering & Technology

Sem: 1st – CSE/CSE(AIML)/DS

Sub: Mathematics-I

Sub Code:BE01R00041

Module 4: Multivariable Calculus (Differentiation)

Weightage: 22%

Topics:

- Function of several variables
- Limit & Continuity
- Partial derivative of higher order
- Mixed derivative theorem
- Chain Rule & Total derivative (Composite function of one & two variables)
- Implicit Function
- Tangent plane & Normal Line
- Extreme values for function of two variables (Maxima , Minima and saddle points)
- Method of Lagrange's Multipliers
- Gradient & Directional derivatives

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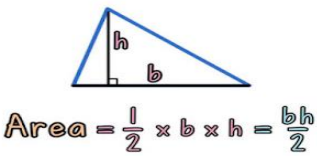
❖ Functions of several variables:

- There are some physical & engineering problems in which the quantities actually depend on more than one independent variables. Thus, the requirements of an extension of a function of more than one variable comes in existence.
- For Example:

  Area of a triangle depends on its base & height hence area is the function of two variables i.e., its base & height

$$u = f(x, y)$$

  u is called a function of two variables x & y ,where x & y are independent variables & u is dependent variable.



- Volume of a rectangular box depends on its length, width & height

$$u = f(x, y, z)$$

u is called a function of three variables x , y & z ,where x, y & z are independent variables & u is dependent variable.

Volume, V = $l\,w\,h$

- Thus, the function of more than one variable.

  i.e., Function of several variables plays an important role in dealing with many practical situations.

❖ Limit of Functions of two variables:

If $f(x, y)$ is a function of two variables x, y then $\lim\limits_{(x,y)\to(a,b)} f(x, y) = l$.

In general, for any chosen number $\epsilon > 0$,if there exists a number $\delta > 0$ such that, for all values of (x, y),

$$|f(x, y) - l| < \epsilon \text{ , whenever } |x - a| < \delta \text{ and } |y - b| < \delta$$

➢ Working Rules to find the limit:

Let $\lim\limits_{(x,y)\to(a,b)} f(x, y)$ be a limit function. Consider the following cases:

- **Step 1:** Find the value of $f(x, y)$ along $y \to b$ and then $x \to a$

  i.e., $\lim\limits_{x\to a}\left\{\lim\limits_{y\to b} f(x, y)\right\} = \lim\limits_{\substack{y\to b\\ x\to a}} f(x, y) = f_1$ (Say)

- **Step 2:** Find the value of $f(x, y)$ along $x \to a$ and then $y \to b$

  i.e., $\lim\limits_{y\to b}\left\{\lim\limits_{x\to a} f(x, y)\right\} = \lim\limits_{\substack{x\to a\\ y\to b}} f(x, y) = f_2$ (Say)

- If both the limits are equal then $\lim\limits_{(x,y)\to(a,b)} f(x, y)$ Exists & that value is the limit of $f(x, y)$ otherwise limit does not exist.

- If $a = 0$ and $b = 0$, Evaluate limit along different paths say

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- **Step 3:** Put $y = mx$ & $x \to 0$

  i.e., $\lim_{x\to 0}\left\{\lim_{y\to mx} f(x,y)\right\} = f_3$ (Say)

  If $f_1 = f_2 \neq f_3$ then limit of a function does not exist.

  If f_3 is a function of m only then limit of a function does not exist.

  If $f_1 = f_2 = f_3$ then go to step 4.

- **Step 4:** Put $y = mx^2$ & $x \to 0$

  i.e., $\lim_{x\to 0}\left\{\lim_{y\to mx^2} f(x,y)\right\} = f_4$ (Say)

  If $f_1 = f_2 = f_3 \neq f_4$ then limit of a function does not exist.

  If f_4 is a function of m only then limit does not exist.

  If $f_1 = f_2 = f_3 = f_4 = l$ (say) then limit of a function exists.

- Thus, $\lim_{(x,y)\to(a,b)} f(x,y) = l$

❖ Continuity:

A function $f(x,y)$ is said to be continuous at a point (a,b) if the following three conditions are satisfied,

1) $f(a,b)$ *exists*.
   $i.e.\, f(x,y)$ is defined at point (a,b).
2) $\lim_{(x,y)\to(a,b)} f(x,y)$ exists.
3) $\lim_{(x,y)\to(0,0)} f(x,y) = f(a,b)$

Note:

i. If $f(x,y)$ & $g(x,y)$ are continuous at (a,b) then
   $f \pm g, f.g, \frac{f}{g} (g \neq 0)$ are continuous at (a,b).

ii. Polynomial & rational functionals are continuous at every point at which they are defined.

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Example 1: Evaluate $\lim_{(x,y)\to(5,-2)}(x^4+4x^3y-5xy^2)$

Solution: Since $f(x,y)=x^4+4x^3y-5xy^2$ is a polynomial

∴ It is continous every where.

So, we can find the limit by direct substitution.

$$\lim_{(x,y)\to(5,-2)}(x^4+4x^3y-5xy^2)=(5)^4+4(5)^3(-2)-5(5)(-2)^2$$

$$=625-1000-100$$

$$=-475$$

Example 2: Evaluate $\lim_{(x,y)\to(0,0)}\frac{x-y+2\sqrt{x}-2\sqrt{y}}{\sqrt{x}-\sqrt{y}}$

Solution: Since denominator $\sqrt{x}-\sqrt{y}$ approaches to 0 $as(x,y)\to(0,0)$,

We get $\left(\frac{0}{0}\right)$ $Form$, So we can't use quotient rule,

$$\lim_{(x,y)\to(0,0)}\frac{x-y+2\sqrt{x}-2\sqrt{y}}{\sqrt{x}-\sqrt{y}}$$

$$=\lim_{(x,y)\to(0,0)}\frac{(\sqrt{x}-\sqrt{y})(\sqrt{x}+\sqrt{y})+2(\sqrt{x}-\sqrt{y})}{(\sqrt{x}-\sqrt{y})}$$

$$=\lim_{(x,y)\to(0,0)}\frac{(\sqrt{x}+\sqrt{y})+2}{1}$$

$$=0+0+2$$

$$=2$$

Example 3: Evaluate $\lim_{(x,y)\to(0,0)}\frac{x^4-y^2}{x^4+y^2}$ **or**

By considering different paths of approach show that $f(x,y)=\frac{x^4-y^2}{x^4+y^2}$ **has no limit as** $(x,y)\to(0,0)$.

Solution:

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- **Step 1:** First $y \to 0$ *then* $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^4-y^2}{x^4+y^2} = \lim_{\substack{y\to 0\\ x\to 0}} \frac{x^4-y^2}{x^4+y^2} = \lim_{x\to 0} \frac{x^4}{x^4} = 1$$

- **Step 2:** First $x \to 0$ *then* $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^4-y^2}{x^4+y^2} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^4-y^2}{x^4+y^2} = \lim_{y\to 0} \frac{-y^2}{y^2} = -1$$

- Both limits are different.

  ∴ Limit does not exists.

Example 4: **Evaluate** $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$

Solution:

- **Step 1:** First $y \to 0$ *then* $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{\substack{y\to 0\\ x\to 0}} \frac{xy}{x^2+y^2} = \lim_{x\to 0} \frac{0}{x^2} = \lim_{x\to 0} 0 = 0$$

- **Step 2:** First $x \to 0$ *then* $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{xy}{x^2+y^2} = \lim_{y\to 0} \frac{0}{y^2} = \lim_{y\to 0} 0 = 0$$

- Both limits are Equal, so apply path-3
- **Step 3:** Put $y = mx$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{x\to 0} \frac{x(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{x^2 m}{x^2+m^2x^2} = \lim_{x\to 0} \frac{m}{1+m^2} = \frac{m}{1+m^2}$$

- Limit depends on m & m is not fixed.

  Thus, the limit of a given function does **not** exist.

Example 5: Evaluate $\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2}$

Solution:

- **Step 1:** First $y \to 0$ *then* $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{\substack{y\to 0\\ x\to 0}} \frac{x^3-y^3}{x^2+y^2} = \lim_{x\to 0} \frac{x^3}{x^2} = \lim_{x\to 0} x = 0$$

- **Step 2:** First $x \to 0$ *then* $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^3-y^3}{x^2+y^2} = \lim_{y\to 0} \frac{-y^3}{y^2} = \lim_{y\to 0} y = 0$$

- Both limits are Equal, so apply path-3

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- **Step 3:** Put $y = mx$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{x\to 0} \frac{x^3-(mx)^3}{x^2+(mx)^2} = \lim_{x\to 0} \frac{x^3-m^3x^3}{x^2+m^2x^2} = \lim_{x\to 0} \frac{x^3(1-m^3)}{x^2(1+m^2)} = 0$$

- **Step 4:** Put $y = mx^2$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{x\to 0} \frac{x^3-(mx^2)^3}{x^2+(mx^2)^2} = \lim_{x\to 0} \frac{x^3-m^3x^6}{x^2+m^2x^4} = \lim_{x\to 0} \frac{x^3(1-m^3x^3)}{x^2(1+m^2x^2)} = 0$$

- Hence, the limit of given function exists.

$$\therefore \lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = 0$$

Example 6: Show that the function $f(x,y) = \frac{2x^2y}{x^4+y^2}$ has no limit as (x, y) approaches to (0,0).

Solution:

- **Step 1:** First $y \to 0$ *then* $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^4+y^2} = \lim_{\substack{y\to 0\\ x\to 0}} \frac{2x^2y}{x^4+y^2} = \lim_{x\to 0} \frac{0}{x^4} = \lim_{x\to 0} 0 = 0$$

- **Step 2:** First $x \to 0$ *then* $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^4+y^2} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{2x^2y}{x^4+y^2} = \lim_{y\to 0} \frac{0}{y^2} = \lim_{y\to 0} 0 = 0$$

- Both limits are Equal, so apply path-3
- **Step 3:** Put $y = mx$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^4+y^2} = \lim_{x\to 0} \frac{2x^2(mx)}{x^4+(mx)^2} = \lim_{x\to 0} \frac{2x^3m}{x^4+m^2x^2} = 0$$

- **Step 4:** Put $y = mx^2$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^4+y^2} = \lim_{x\to 0} \frac{2x^2(mx^2)}{x^4+(mx^2)^2} = \lim_{x\to 0} \frac{2x^4m}{x^4+m^2x^4} = \frac{2m}{1+m^2}$$

- Limit depends on m & m is not fixed.

  Thus, the limit of a given function does **not** exist.

Example 7: Determine whether $\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^3+y^3}$, exist and find it if exists.

Solution:

- **Step 1:** First $y \to 0$ *then* $x \to 0$

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$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^3+y^3} = \lim_{\substack{y\to 0\\ x\to 0}} \frac{2x^2y}{x^3+y^3} = \lim_{x\to 0} \frac{0}{x^3} = \lim_{x\to 0} 0 = 0$$

- **Step 2:** First $x \to 0$ *then* $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^3+y^3} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{2x^2y}{x^3+y^3} = \lim_{y\to 0} \frac{0}{y^3} = \lim_{y\to 0} 0 = 0$$

Both limits are Equal, so apply path-3

- **Step 3:** Put $y = mx$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^3+y^3} = \lim_{x\to 0} \frac{2x^2(mx)}{x^3+(mx)^3} = \lim_{x\to 0} \frac{2mx^3}{x^3+m^2x^3} = \frac{2m}{1+m^3}$$

- Limit depends on m & m is not fixed.

Thus, the limit of a given function does **not** exist.

H.W.

1) **Evaluate** $\lim_{(x,y)\to(0,1)} \frac{x-xy+3}{x^2y+5xy-y^3}$ **Ans:** -3

2) **Evaluate** $\lim_{(x,y)\to(0,0)} \frac{x^2-xy}{\sqrt{x}-\sqrt{y}}$ **Ans:** 0

3) **Evaluate** $\lim_{(x,y)\to(0,0)} \frac{x-y}{x+y}$ **Ans:** Limit does not exist.

4) **Evaluate** $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$ **Ans:** Limit does not exist.

5) **Show that function** $f(x,y) = \frac{2x^2y^2}{x^4+y^4}$ **has no limit as** (x,y) **approaches to** $(0,0)$.

6) **Show that the function** $f(x,y) = \frac{2x^2y^2}{x^4+y^4}$ **has no limit as** (x, y) **approaches to OR**

Show that following limit does not exist using different path approach $\lim_{(x,y)\to(0,0)} \frac{2x^2y^2}{x^4+y^4}$

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Examples of Continuity:

Example 1: Show that $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} & ; \quad (x,y) \neq (0,0) \\ 0 & ; \quad (x,y) = (0,0) \end{cases}$ **is continuous at every point except at the origin.**

Solution:

For $(x,y) \neq (0,0)$ **,** Function is rational function. So, it is continuous.

But for $(x,y) = (0,0)$

(1) $f(0,0) = 0$

(2) **Limit Existence:**

- **Step 1:** First $y \to 0$ $then$ $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2xy}{x^2+y^2} = \lim_{\substack{y\to 0 \\ x\to 0}} \frac{2xy}{x^2+y^2} = \lim_{x\to 0} \frac{0}{x^2} = \lim_{x\to 0} 0 = 0$$

- **Step 2:** First $x \to 0$ $then$ $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2xy}{x^2+y^2} = \lim_{\substack{x\to 0 \\ y\to 0}} \frac{2xy}{x^2+y^2} = \lim_{y\to 0} \frac{0}{y^2} = \lim_{y\to 0} 0 = 0$$

- Both limits are Equal, so apply path-3
- **Step 3:** Put $y = mx$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2xy}{x^2+y^2} = \lim_{x\to 0} \frac{2x(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{2mx^2}{x^2+m^2x^2} = \frac{2m}{1+m^2}$$

Limit depends on m & m is not fixed.

Thus, the limit of a given function does **not** exist.

➢ Hence, the given function is **not** continuous at origin $(0,0)$.

Example 2: Show that $f(x,y) = \begin{cases} \frac{2x^2y}{x^3+y^3} & ; \quad (x,y) \neq (0,0) \\ 0 & ; \quad (x,y) = (0,0) \end{cases}$ **is not continuous at the origin.**

Solution:

For $(x,y) \neq (0,0)$ **,** Function is rational function. So, it is continuous.

But for $(x,y) = (0,0)$

(1) $f(0,0) = 0$

(2) **Limit Existence:**

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- **Step 1:** First $y \to 0$ $then$ $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^3+y^3} = \lim_{\substack{y\to 0\\ x\to 0}} \frac{2x^2y}{x^3+y^3} = \lim_{x\to 0} \frac{0}{x^3} = \lim_{x\to 0} 0 = 0$$

- **Step 2:** First $x \to 0$ $then$ $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^3+y^3} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{2x^2y}{x^3+y^3} = \lim_{y\to 0} \frac{0}{y^3} = \lim_{y\to 0} 0 = 0$$

- Both limits are Equal, so apply path-3
- **Step 3:** Put $y = mx$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^3+y^3} = \lim_{x\to 0} \frac{2x^2(mx)}{x^3+(mx)^3} = \lim_{x\to 0} \frac{2mx^3}{x^3+m^2x^3} = \frac{2m}{1+m^3}$$

Limit depends on m & m is not fixed.

Thus, the limit of a given function does **not** exist.

➢ Hence, the given function is **not** continuous at origin $(0,0)$.

Example 3: Show that $f(x,y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & ; \quad (x,y) \neq (0,0) \\ 0 & ; \quad (x,y) = (0,0) \end{cases}$ is continuous at the origin.

Solution:

For $(x,y) \neq (0,0)$ **,** Function is rational function. So, it is continuous.

But for $(x,y) = (0,0)$

(1) $f(0,0) = 0$

(2) **Limit Existence:**

- **Step 1:** First $y \to 0$ $then$ $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = \lim_{\substack{y\to 0\\ x\to 0}} \frac{3x^2y}{x^2+y^2} = \lim_{x\to 0} \frac{0}{x^2} = \lim_{x\to 0} 0 = 0$$

- **Step 2:** First $x \to 0$ $then$ $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{3x^2y}{x^2+y^2} = \lim_{y\to 0} \frac{0}{y^2} = \lim_{y\to 0} 0 = 0$$

- Both limits are Equal, so apply path-3
- **Step 3:** Put $y = mx$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = \lim_{x\to 0} \frac{3x^2(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{3mx^3}{x^2+m^2x^2} = \lim_{x\to 0} x\left(\frac{3m}{1+m^2}\right) = 0$$

- **Step 4:** Put $y = mx^2$ & $x \to 0$

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$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = \lim_{x\to 0} \frac{3x^2(mx^2)}{x^2+(mx^2)^2} = \lim_{x\to 0} \frac{3mx^4}{x^2+m^2x^4} = \lim_{x\to 0} x^2\left(\frac{3m}{1+m^2x^2}\right) = 0$$

$\therefore$ limit exists & equal to 0.

$$\therefore \lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0$$

(3) $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0) = 0$

All three conditions of continuity are satisfied.

- Hence, $f(x,y)$ is continuous at origin $(0,0)$.

Example 4: Discuss the continuity of the given function $\boldsymbol{f(x,y) = \begin{cases} \dfrac{x^3-y^3}{x^2+y^2} & ; x \neq 0, y \neq 0 \\ 0 & ; x = 0, y = 0 \end{cases}}$

Solution:

For $(x,y) \neq (0,0)$ **,** Function is rational function. So, it is continuous.

But for $(x,y) = (0,0)$

(1) $f(0,0) = 0$

(2) **Limit Existence:**

- **Step 1:** First $y \to 0$ *then* $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{\substack{y\to 0\\ x\to 0}} \frac{x^3-y^3}{x^2+y^2} = \lim_{x\to 0} \frac{x^3}{x^2} = \lim_{x\to 0} x = 0$$

- **Step 2:** First $x \to 0$ *then* $y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^3-y^3}{x^2+y^2} = \lim_{y\to 0} \frac{-y^3}{y^2} = \lim_{y\to 0} y = 0$$

- Both limits are Equal, so apply path-3
- **Step 3:** Put $y = mx$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{x\to 0} \frac{x^3-(mx)^3}{x^2+(mx)^2} = \lim_{x\to 0} \frac{x^3-m^3x^3}{x^2+m^2x^2} = \lim_{x\to 0} \frac{x^3(1-m^3)}{x^2(1+m^2)} = 0$$

- **Step 4:** Put $y = mx^2$ & $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{x\to 0} \frac{x^3-(mx^2)^3}{x^2+(mx^2)^2} = \lim_{x\to 0} \frac{x^3-m^3x^6}{x^2+m^2x^4} = \lim_{x\to 0} \frac{x^3(1-m^3x^3)}{x^2(1+m^2x^2)} = 0$$

- Hence, the limit of given function exists.

  $\therefore$ limit exists & equal to 0.

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$$\therefore \lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2} = 0$$

(3) $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0) = 0$

All three conditions of continuity are satisfied.

- Hence, $f(x,y)$ is continuous at origin $(0,0)$.

Example 5: Discuss continuity of $\boldsymbol{f(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2}, & for\ (x,y) \neq (0,0) \\ 0, & for\ (x,y) = (0,0) \end{cases}\ in\ R \times R}$

Solution:

For $(x,y) \neq (0,0)$ **,** Function is rational function. So, it is continuous.

But for $(x,y) = (0,0)$

(1) $f(0,0) = 0$

(2) **Limit Existence:**

- **Step 1:** First $y \to 0\ then\ x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} = \lim_{\substack{y\to 0 \\ x\to 0}} \frac{x^2-y^2}{x^2+y^2} = \lim_{x\to 0} \frac{x^2}{x^2} = 1$$

- **Step 2:** First $x \to 0\ then\ y \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} = \lim_{\substack{x\to 0 \\ y\to 0}} \frac{x^2-y^2}{x^2+y^2} = \lim_{y\to 0} \frac{-y^2}{y^2} = -1$$

- Both limits are different.

  $\therefore$ Limit does not exists.

- Hence, the given function is **not** continuous at origin $(0,0)$.

H.W.

1) **Show that** $\boldsymbol{f(x,y) = \begin{cases} \frac{x^3y}{x^4+y^4} & ; \quad (x,y) \neq (0,0) \\ 0 & ; \quad (x,y) = (0,0) \end{cases}}$ **is not continuous at the origin.**

2) **Show that** $\boldsymbol{f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & ; \quad (x,y) \neq (0,0) \\ 0 & ; \quad (x,y) = (0,0) \end{cases}}$ **is continuous at the origin.**

3) **Discuss the continuity of** $\boldsymbol{f(x,y) = \begin{cases} \frac{x}{\sqrt{x^2+y^2}}, & x \neq 0, y \neq 0 \\ 2, & x = 0, y = 0 \end{cases}}$ **at the origin.**

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❖ Partial Derivatives:

- For the function $y = f(x)$, derivative of y $w.r.t$ $x, denoted\ by\ \frac{dy}{dx}\ or\ y'\ or\ y_1$

  This is called Ordinary derivative as we differentiate the function of one independent variable only.

- When the function involves two or more independent variable like

$$u = f(x, y)$$

  Then derivative of u $w.r.t.$ any one of the independent variables treating all other variables as constant is known as Partial Derivatives of u $w.r.t.$ that variable.

- Partial Derivative of u with respect to x, keeping y as a constant is called the first order partial derivative of u $w.r.t$ x and is denoted by $\frac{\partial u}{\partial x}$ or $\frac{\partial f}{\partial x}$ or f_x or u_x

$$\frac{\partial u}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

- Similarly, Partial Derivative of u with respect to y, keeping x as a constant is called the first order partial derivative of u $w.r.t$ y and is denoted by $\frac{\partial u}{\partial y}$ or $\frac{\partial f}{\partial y}$ or f_y or u_y

$$\frac{\partial u}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\partial y}$$

❖ Higher Order Partial Derivatives:

Partial derivatives of higher order of a function $u = f(x, y)$ are obtained by partial differentiation of first order partial derivative. Thus, if $u = f(x, y)$ then

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = f_{xx}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = f_{yy}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = f_{yx}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = f_{xy}$$

are called second order partial derivatives. Similarly, other higher order derivatives can also be obtained.

Remarks: f_{xy} & f_{yx} are called mixed or crossed partial derivatives of f.

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❖ Clairaut's Theorem or The Mixed Derivative Theorem:

If $u = f(x, y)$ possesses continuous second order partial derivatives

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} \Longrightarrow f_{xy} = f_{yx}$$

This is also called commutative property.

Example 1: If $f(x, y) = x^2y + xy^2$ then find $f_x(1, 2)$ and $f_y(1, 2)$ by definition.

Solution:

- $\left(\frac{\partial f}{\partial x}\right)_{(x_0,y_0)} = \lim_{\Delta x \to 0} \frac{f(x_0+\Delta x, y_0) - f(x_0, y_0)}{\Delta x}$
- $\left(\frac{\partial f}{\partial x}\right)_{(1,2)} = \lim_{\Delta x \to 0} \frac{f(1+\Delta x, 2) - f(1,2)}{\Delta x}$

$$= \lim_{\Delta x \to 0} \frac{(1+\Delta x)^2 2 + (1+\Delta x)4 - 6}{\Delta x}$$

$$= \lim_{\Delta x \to 0} 2\left[\frac{(1+\Delta x)^2 + (1+\Delta x)2 - 3}{\Delta x}\right]$$

$$= \lim_{\Delta x \to 0} 2\left[\frac{1 + 2\Delta x + (\Delta x)^2 + 2 + 2\Delta x - 3}{\Delta x}\right]$$

$$= \lim_{\Delta x \to 0} 2\left[\frac{4\Delta x + (\Delta x)^2}{\Delta x}\right]$$

$$= \lim_{\Delta x \to 0} 2[4 + \Delta x]$$

$$= \lim_{\Delta x \to 0} [8 + 2\Delta x]$$

$$= 8$$

- $\left(\frac{\partial f}{\partial y}\right)_{(x_0,y_0)} = \lim_{\Delta y \to 0} \frac{f(x_0, y_0+\Delta y) - f(x_0, y_0)}{\Delta x}$
- $\left(\frac{\partial f}{\partial y}\right)_{(1,2)} = \lim_{\Delta y \to 0} \frac{f(1, 2+\Delta y) - f(1,2)}{\Delta x}$

$$= \lim_{\Delta y \to 0} \frac{(2+\Delta y)^2 1 + (2+\Delta y)1 - 6}{\Delta y}$$

$$= \lim_{\Delta y \to 0} \left[\frac{2 + \Delta y + 4 + 4\Delta y + (\Delta y)^2 - 6}{\Delta y}\right]$$

$$= \lim_{\Delta y \to 0} \left[\frac{5\Delta y + (\Delta y)^2}{\Delta y}\right]$$

$$= \lim_{\Delta y \to 0} [5 + \Delta y]$$

$$= 5$$

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Example 2: Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial^2 f}{\partial x^2}$ & $\frac{\partial^2 f}{\partial x \partial y}$ at point $(1, 2)$ where $f(x, y) = x^2 + 3xy + y - 1$.

Solution:

- Here, $f(x, y) = x^2 + 3xy + y - 1$
- Differentiating f partially with respect to x, we get

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 + 3xy + y - 1) = 2x + 3y \quad \ldots\ldots(1)$$

$$\therefore \left(\frac{\partial f}{\partial x}\right)_{(1,-2)} = 2(1) + 3(2) = 2 + 6 = 8$$

- Now, differentiating f partially with respect to y, we get

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2 + 3xy + y - 1) = 3x + 1 \quad \ldots\ldots(2)$$

$$\therefore \left(\frac{\partial f}{\partial y}\right)_{(1,2)} = 3 + 1 = 4$$

- Differentiating $Eq.(1)$ partially with respect to x, we get

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(2x + 3y) = 2$$

$$\therefore \left(\frac{\partial^2 f}{\partial x^2}\right)_{(1,2)} = 2$$

- Differentiating $Eq.(2)$ partially with respect to x, we get

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(3x + 1) = 3$$

$$\therefore \left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(1,2)} = 3$$

Example 3: If $u = \log(tanx + tany)$ then show that $sin2x\frac{\partial u}{\partial x} + sin2y\frac{\partial u}{\partial y} = 2$

Solution: we have $u = \log(tanx + tany)$

- Differentiating u partially w.r.t. x, y

$$\frac{\partial u}{\partial x} = \frac{1}{tanx + tany}(sec^2 x)$$

$$\frac{\partial u}{\partial y} = \frac{1}{tanx + tany}(sec^2 y)$$

$$\therefore sin2x\frac{\partial u}{\partial x} + sin2y\frac{\partial u}{\partial y}$$

$$= sin2x\frac{1}{tanx + tany}(sec^2 x) + sin2y\frac{1}{tanx + tany}(sec^2 y)$$

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$$= 2sinx.cosx\frac{sec^2x}{tanx+tany} + 2siny.cosy\ \frac{sec^2y}{tanx+tany}$$

$$= \frac{2\frac{sinx}{cosx}}{tanx+tany} + \frac{2\frac{siny}{cosy}}{tanx+tany}$$

$$= \frac{2tanx}{tanx+tany} + \frac{2tany}{tanx+tany}$$

$$= \frac{2(tanx+tany)}{(tanx+tany)}$$

$$= 2$$

Example 4: If $u = \frac{e^{x+y+z}}{e^x+e^y+e^z}$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 2u$

Solution: we have $u = \frac{e^{x+y+z}}{e^x+e^y+e^z}$

- Differentiating u partially w.r.to x,

$$\frac{\partial u}{\partial x} = \left(\frac{(e^x+e^y+e^z)(e^{x+y+z}) - e^{x+y+z}(e^x)}{(e^x+e^y+e^z)^2}\right)$$

$$= \frac{(e^{x+y+z})}{e^x+e^y+e^z} - \frac{e^{x+y+z}(e^x)}{(e^x+e^y+e^z)^2}$$

$$= \frac{(e^{x+y+z})}{e^x+e^y+e^z}\left[1 - \frac{(e^x)}{e^x+e^y+e^z}\right]$$

- Differentiating u partially w.r.to y,

$$\frac{\partial u}{\partial y} = \frac{(e^{x+y+z})}{e^x+e^y+e^z}\left[1 - \frac{(e^y)}{e^x+e^y+e^z}\right]$$

- Differentiating u partially w.r.to z,

$$\frac{\partial u}{\partial z} = \frac{(e^{x+y+z})}{e^x+e^y+e^z}\left[1 - \frac{(e^z)}{e^x+e^y+e^z}\right]$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{(e^{x+y+z})}{(e^x+e^y+e^z)^1}\left[3 - \frac{e^x+e^y+e^z}{e^x+e^y+e^z}\right](3-1) = 2u$$

Example 5: If $u = x^3y + e^{xy^2}$, show that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

Solution: Here, $u = x^3y + e^{xy^2}$

- Differentiating u partially with respect to x, we get

$$\frac{\partial u}{\partial x} = 3x^2y + e^{xy^2}y^2$$

- Again differentiating $\frac{\partial u}{\partial x}$ partially with respect to y, we get

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$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = 3x^2 + y^2 e^{xy^2}(2xy) + (2y)e^{xy^2}$$

$$= 3x^2 + 2ye^{xy^2}(xy^2 + 1) \quad \text{---------- (1)}$$

- Now, differentiating u with respect to y, we get

$$\frac{\partial u}{\partial y} = x^3 + e^{xy^2}(2xy)$$

- Again differentiating $\frac{\partial u}{\partial y}$ with respect to x, we get

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = 3x^2 + e^{xy^2}(y^2)2xy + e^{xy^2}(2y)$$

$$= 3x^2 + 2ye^{xy^2}(xy^2 + 1) \text{ ---------- (2)}$$

- From (1) and (2), we get

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

Example 6: Prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$ for $z = x + y^x$

Solution: we have $z = x + y^x$.

- Differentiating z partially w.r.to x,

$$\frac{\partial z}{\partial x} = 1 + y^x \log y$$

- Differentiating z partially w.r.to y,

$$\frac{\partial z}{\partial y} = 0 + xy^{x-1}$$

- Differentiating $\frac{\partial z}{\partial x}$ partially w.r.to y,

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}(1 + y^x \log y)$$

$$= 0 + y^x\left(\frac{1}{y}\right) + \log y \ (xy^{x-1})$$

$$= y^{x-1}(x \log y + 1) \ldots \ldots (1)$$

- Differentiating $\frac{\partial z}{\partial y}$ partially w.r.to x,

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$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}(xy^{x-1})$$

$$= x(y^{x-1}\log y) + (1)y^{x-1}$$

$$= y^{x-1}(x\log y + 1) \dots\dots (2)$$

- From equation (1) and (2) , $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$

Example 7: Prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, and $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ for $u = tan^{-1}\left(\frac{y}{x}\right)$.

Solution: we have $u = tan^{-1}\left(\frac{y}{x}\right)$

- Differentiating u partially w.r.to x

$$\frac{\partial u}{\partial x} = \frac{1}{1+\left(\frac{y}{x}\right)^2}\left(\frac{-y}{x^2}\right) = \frac{-y}{x^2+y^2}$$

- Differentiating u partially w.r.to y

$$\frac{\partial u}{\partial y} = \frac{1}{1+\left(\frac{y}{x}\right)^2}\left(\frac{1}{x}\right) = \frac{x}{x^2+y^2}$$

- Differentiating $\frac{\partial u}{\partial x}$ partially w.r.to x

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{-y}{x^2+y^2}\right)$$

$$= -y\frac{\partial}{\partial x}\left(\frac{1}{x^2+y^2}\right)$$

$$= -y\left[\frac{-2x}{(x^2+y^2)^2}\right]$$

$$= \frac{2xy}{(x^2+y^2)^2} \dots\dots (1)$$

- Differentiating $\frac{\partial u}{\partial y}$ partially w.r.to y

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{x}{x^2+y^2}\right)$$

$$= x\frac{\partial}{\partial y}\left(\frac{1}{x^2+y^2}\right)$$

$$= x\left[\frac{-2y}{(x^2+y^2)^2}\right]$$

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$$= \frac{-2xy}{(x^2+y^2)^2} \dots\dots (2)$$

- From (1) & (2), $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2xy}{(x^2+y^2)^2} - \frac{2xy}{(x^2+y^2)^2} = 0$
- Differentiating $\frac{\partial u}{\partial x}$ partially w.r.to y

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial y}\left(\frac{-y}{x^2+y^2}\right)$$

$$= \frac{(x^2+y^2)(-1)-(-y)(2y)}{(x^2+y^2)^2}$$

$$= \frac{(y^2-x^2)}{(x^2+y^2)^2} \dots\dots (3)$$

- Differentiating $\frac{\partial u}{\partial y}$ partially w.r.to x

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right)$$

$$= \frac{(x^2+y^2)(1)-(x)(2x)}{(x^2+y^2)^2}$$

$$= \frac{(y^2-x^2)}{(x^2+y^2)^2} \dots\dots (4)$$

- From (3) & (4), $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$

Example 8: If $u = e^{3xyz}$ then show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (3 + 27xyz + 27x^2y^2z^2)e^{3xyz}$

Solution: We have $u = e^{3xyz}$

- Differentiating u partially w.r.to z

$$\frac{\partial u}{\partial z} = e^{3xyz}(3xy)$$

- Differentiating $\frac{\partial u}{\partial z}$ partially w.r.to y

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial z}\right) = \frac{\partial}{\partial y}[e^{3xyz}(3xy)]$$

$$= e^{3xyz}(3x) + (3xy)e^{3xyz}(3xz)$$

$$= e^{3xyz}(3x + 9x^2yz)$$

- Differentiating $\frac{\partial^2 u}{\partial y \partial z}$ partially w.r.to x

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = \frac{\partial}{\partial x}\left(\frac{\partial^2 u}{\partial y \partial z}\right)$$

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$$= \frac{\partial}{\partial x}[e^{3xyz}(3x + 9x^2yz)]$$
$$= (e^{3xyz})(3 + 18xyz) + (e^{3xyz})(3yz)(3x + 9x^2yz)$$
$$= (3 + 18xyz + 9xyz + 27x^2y^2z^2)(e^{3xyz})$$
$$= (3 + 27xyz + 27x^2y^2z^2)(e^{3xyz})$$

Example 9: If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$

Solution: Here, $u = \log(x^3 + y^3 + z^3 - 3xyz)$

- Differentiating u partially with respect to x, we get

$$\therefore \frac{\partial u}{\partial x} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz}$$

Similarly,

$$\frac{\partial u}{\partial y} = \frac{3y^2-3xz}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial z} = \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz}$$

- $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3x^2+3y^2+3z^2-3xy-3yz-3xz}{x^3+y^3+z^3-3xyz}$

$$= \frac{3(x^2+y^2+z^2-xy-yz-zx)}{(x+y+z)(x^2+y^2+z^2-xy-yz-zx)}$$

$$= \frac{3}{x+y+z} \quad \ldots\ldots(1)$$

- $L.H.S. = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)u$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}\right)$$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)\left(\frac{3}{x+y+z}\right) \qquad \{From\ Eq(1)\}$$

$$= \frac{\partial}{\partial x}\left(\frac{3}{x+y+z}\right) + \frac{\partial}{\partial y}\left(\frac{3}{x+y+z}\right) + \frac{\partial}{\partial z}\left(\frac{3}{x+y+z}\right)$$

$$= -\frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2}$$

$$= -\frac{9}{(x+y+z)^2} = R.H.S$$

- Hence, $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$

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Example 10: If $u = f(x+at) + g(x-at)$, prove that $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$

Solution: Here, $u = f(x+at) + g(x-at)$

- Differentiating u partially with respect to t , we get

$$\frac{\partial u}{\partial t} = f'(x+at)(a) + g'(x-at)(-a)$$

$$= af'(x+at) - ag'(x-at)$$

- Again differentiating $\frac{\partial u}{\partial t}$ partially with respect to t, we get

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial t}\right) = af''(x+at)(a) - ag''(x-at)(-a)$$

$$= a^2 f''(x+at) + a^2 g''(x-at) \text{ ---------- } (1)$$

- Now, differentiating u partially with respect to x, we get

$$\frac{\partial u}{\partial x} = f'(x+at)(1) + g'(x-at)(1)$$

$$= f'(x+at) + g'(x-at)$$

- Again differentiating $\frac{\partial u}{\partial x}$ partially with respect to x, we get

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = f''(x+at)(1) + g''(x-at)(1)$$

$$= f''(x+at) + g''(x-at)$$

- $a^2 \frac{\partial^2 u}{\partial x^2} = a^2[f''(x+at) + g''(x-at)]$

$$= a^2 f''(x+at) + a^2 g''(x-at) \text{ ---------- } (2)$$

- From (1) and (2), we get $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$

H.W.

1) Let $w = f(x,y,z)$ be a function of three independent variables. Write the normal definition of the partial derivative for $\frac{\partial f}{\partial z}$ at point (x_0, y_0, z_0), Using this definition find $\frac{\partial f}{\partial z}$ at (1,2,3) for $f(x,y,z) = x^2yz^2$ **Ans:** 12

2) If $u = x^2y + y^2z + z^2x$,find the value of

a) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ **Ans:** $(x+y+z)^2$

b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$ **Ans:** $2(x+y+z)$

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c) $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u$ **Ans:** $6(x+y+z)$

3) $u = z\,tan^{-1}\left(\frac{y}{x}\right)$ then prove that $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=0$

4) If $u=\log(x^2+y^2)$ then show that $(i)\ \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0\ (ii)\ \frac{\partial^2 u}{\partial x \partial y}=\frac{\partial^2 u}{\partial y \partial x}$

5) If $u=e^{x^y}$ then find $\frac{\partial^2 u}{\partial y \partial x}$ **Ans:** $= x^{y-1}e^{x^y}(yx^y\log x+1+y\log x)$

6) If $\theta = t^n e^{-\frac{r^2}{4t}}$ then find n so that $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\theta}{\partial r}\right)=\frac{\partial\theta}{\partial t}$ **Ans:** $\left(-\frac{3}{2}\right)$

❖ Differentiation of a function of a function (Chain Rule):

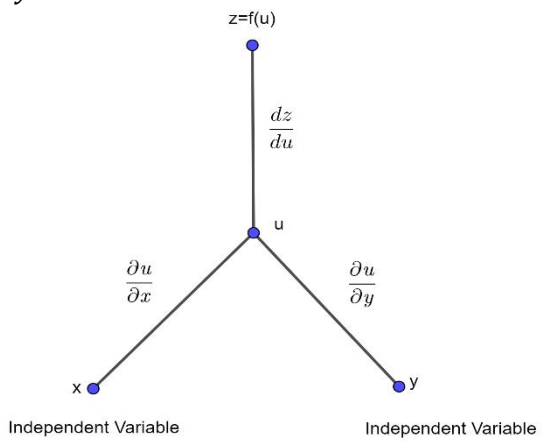
- Let $z=f(u)$ & $u=\emptyset(x,y)$

  $z \to u \to x, y$

  $i.e$ z is a function of u & u itself is a function of two independent variables x & y.

  Thus, z is called a function of a function of x & y.

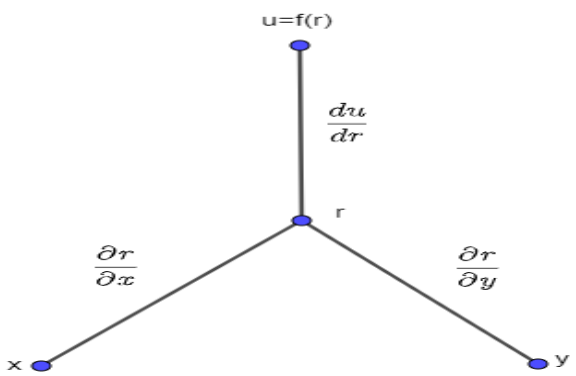
- $\frac{\partial z}{\partial x}=\frac{dz}{du}.\frac{\partial u}{\partial x}=f'(u).\frac{\partial u}{\partial x}$
- $\frac{\partial z}{\partial y}=\frac{dz}{du}.\frac{\partial u}{\partial y}=f'(u).\frac{\partial u}{\partial y}$



Tree Diagram

Example 1: If $u=f(r)$ and $r^2=x^2+y^2$, prove that $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=f''(r)+\frac{1}{r}f'(r)$

Solution: Here, $u=f(r), r^2=x^2+y^2$



Tree Diagram

- Firstly, differentiating r partially with respect to x, yrespectively, we get

$$\Rightarrow 2r\frac{\partial r}{\partial x} = 2x$$

$$\Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r} \qquad \text{Similarly, } \frac{\partial r}{\partial y} = \frac{y}{r}$$

- Differentiating u partially with respect to x , we get

$$\frac{\partial u}{\partial x} = \frac{du}{dr}\cdot\frac{\partial r}{\partial x} = f'(r)\cdot\frac{\partial r}{\partial x} = f'(r).\frac{x}{r}$$

- Again differentiating $\frac{\partial u}{\partial x}$ partially with respect to x, we get

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)$$

$$= \frac{\partial}{\partial x}\left[f'(r)\cdot\frac{x}{r}\right]$$

$$= \frac{x}{r}\frac{\partial f'(r)}{\partial x} + f'(r)\frac{1}{r} + f'(r)x\frac{\partial}{\partial x}\left(\frac{1}{r}\right)$$

$$= \frac{x}{r}f''(r)\frac{\partial r}{\partial x} + f'(r)\frac{1}{r} + f'(r)x\left(-\frac{1}{r^2}\cdot\frac{\partial r}{\partial x}\right)$$

$$= \frac{x}{r}f''(r)\frac{x}{r} + f'(r)\frac{1}{r} - f'(r)x\frac{x}{r^3}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{x^2}{r^2}f''(r) + f'(r)\frac{1}{r} - \frac{x^2}{r^3}f'(r) \qquad ---------- (1)$$

Similarly, we get

$$\frac{\partial^2 u}{\partial y^2} = \frac{y^2}{r^2}f''(r) + f'(r)\frac{1}{r} - \frac{y^2}{r^3}f'(r) \qquad ---------- (2)$$

- From (1), (2) and (3), we get

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = (x^2+y^2)\frac{f''(r)}{r^2} + 2f'(r)\frac{1}{r} - (x^2+y^2)\frac{f'(r)}{r^3}$$

$$= r^2\frac{f''(r)}{r^2} + 2f'(r)\frac{1}{r} - r^2\frac{f'(r)}{r^3} \quad (\because r^2 = x^2+y^2)$$

$$= f''(r) + 2\frac{f'(r)}{r} - \frac{f'(r)}{r}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{f'(r)}{r}$$

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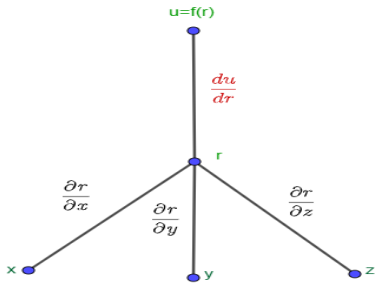
Example 2: If $u = r^m$ and $r = \sqrt{x^2 + y^2 + z^2}$,

prove that $u_{xx}+u_{yy}+u_{zz} = m(m+1)r^{m-2}$.

Solution: Here, $u = f(r), r^2 = x^2 + y^2 + z^2$



Tree Diagram

- Firstly, differentiating r partially with respect to x, y & z respectively, we get

$$\Rightarrow 2r\frac{\partial r}{\partial x} = 2x$$

$$\Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r} \qquad \text{Similarly, } \frac{\partial r}{\partial y} = \frac{y}{r} \ \& \ \frac{\partial r}{\partial z} = \frac{z}{r}$$

- Differentiating u partially with respect to x , we get

$$\frac{\partial u}{\partial x} = \frac{du}{dr}\cdot\frac{\partial r}{\partial x} = f'(r)\cdot\frac{\partial r}{\partial x} = m.r^{m-1}.\frac{x}{r} = m.r^{m-2}.x$$

- Again differentiating $\frac{\partial u}{\partial x}$ partially with respect to x, we get

$$\Rightarrow u_{xx} = \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)$$

$$= \frac{\partial}{\partial x}[m.r^{m-2}.x\,]$$

$$= m\frac{\partial}{\partial x}[r^{m-2}.x\,]$$

$$= m\left[r^{m-2}(1) + \left\{(m-2)r^{m-3}\frac{\partial r}{\partial x}\right\}x\right]$$

$$= m\left[r^{m-2} + (m-2)r^{m-3}\frac{x}{r}x\right]$$

$$= m[r^{m-2} + (m-2)r^{m-4}x^2]$$

$$\Rightarrow u_{xx} = \frac{\partial^2 u}{\partial x^2} = m[r^{m-2} + (m-2)r^{m-4}x^2] \qquad ---------- (1)$$

Similarly, we get

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$$\Rightarrow u_{yy} = \frac{\partial^2 u}{\partial y^2} = m[r^{m-2} + (m-2)r^{m-4}y^2] \qquad ---------(2)$$

$$\Rightarrow u_{zz} = \frac{\partial^2 u}{\partial z^2} = m[r^{m-2} + (m-2)r^{m-4}z^2] \qquad ---------(3)$$

- From (1), (2) and (3), we get

$$\therefore u_{xx}+u_{yy}+u_{zz} = 3mr^{m-2} + m(m-2)r^{m-4}(x^2+y^2+z^2)$$

$$= 3mr^{m-2} + m(m-2)r^{m-4}(r^2) \quad (\because r^2 = x^2+y^2+z^2)$$

$$= 3mr^{m-2} + m(m-2)r^{m-2}$$

$$= mr^{m-2}(3+m-2)$$

$$\therefore u_{xx}+u_{yy}+u_{zz} = m(m+1)r^{m-2}$$

H.W.

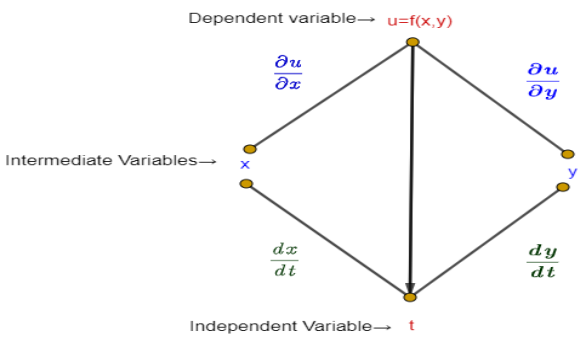
1) If $u = f(r)$ and $r^2 = x^2+y^2+z^2$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r}f'(r)$

2) If $x = rcos\theta$, $y = rsin\theta$ then prove that $\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2 = 1$

❖ Total Derivative:

Composite function of one variable:

- If $u = f(x,y)$, where $x = \phi(t)$ and $y = \Psi(t)$ then we can express u as function of t then derivative of u $w.r.t.$ t is given by the chain rule,
- $\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt}$ …… (1) is called **Total Derivative of u**.



- This equation (1) can be written in terms of differential as

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy \ldots\ldots\ldots (2)$$

is called **Total Differential of u** *or* **ExactDifferential of u**

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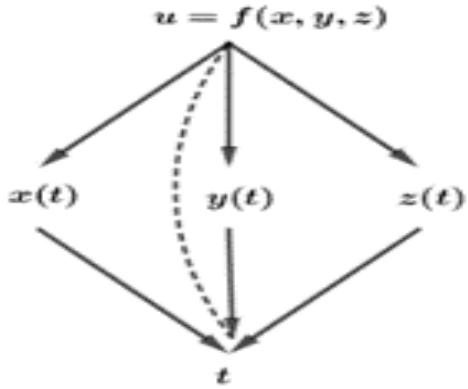
- If we put $t = x$ in Eq(1),

  $\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dx}$ …… (3)

  is called the **Total Differential coefficient of u w.r.t. x**

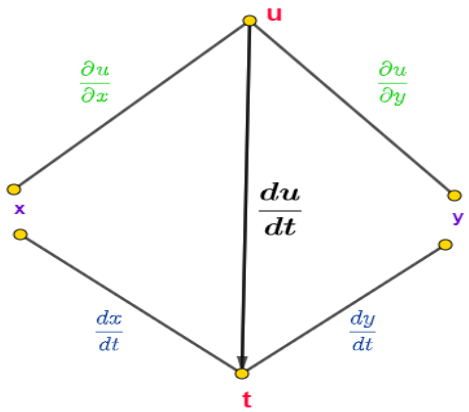
- If $u = f(x, y, z)$ where $x = \emptyset(t), y = \varphi(t), z = \phi(t)$ then total derivative of u is given as

$$\frac{du}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} + \frac{\partial z}{\partial z}\frac{dz}{dt}$$



Example 1: If $u = y^2 - 4ax, x = at^2, y = 2at$ find $\frac{du}{dt}$

Solution: $u \to x, y \to t$



- $\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$

  $= (-4a)(2at) + (2y)(2a)$

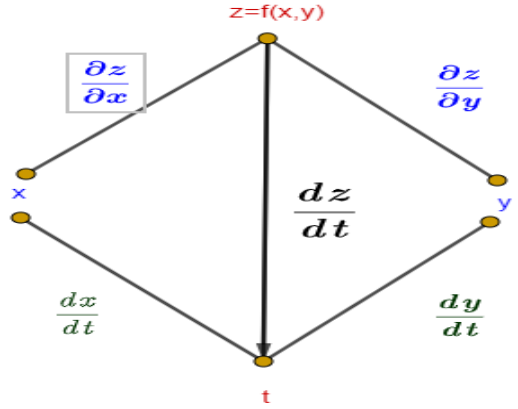
  $= (-8a^2t) + 2(2at)(2a)$

  $= (-8a^2t) + (8a^2t)$

$\therefore \frac{du}{dt} = 0$

Example 2: If $z = x^2y + 3xy^4$ where $x = sin2t$ and $y = cost$ find $\frac{dz}{dt}$ when $t = 0$

Solution: $z \to x, y \to t$



- $\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$

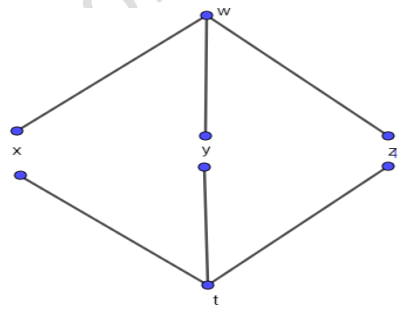
$$= (2xy + 3y^4)(2cos2t) + (x^2 + 12xy^3)(-sint)$$

$$= (2sin2t.cost + 3cos^4t)(2cos2t) + (sin^2 2t + 12sin2tcos^3t)(-sint)$$

$$\therefore \left(\frac{dz}{dt}\right)_{t=0} = [2(0)(1) + 3(1)]2(1) + (0)(0) = 6$$

Example 3: Find $\frac{dw}{dt}$, if $w = xy + z$; where $x = cost, y = sint, z = t$

Solution: $w \to x, y, z \to t$



- $\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$

$$= (y)(-sint) + (x)(cost) + (1)(1)$$

$$= (sint)(-sint) + (cost)(cost) + (1)(1)$$

$$= -sin^2t + cos^2t + 1 = 1 + cos2t$$

Example 4: Find $\frac{dw}{dt}$, if $w = xy^2 + z^2$; where $x = cost, y = sint, z = t$

Solution: $w \to x, y, z \to t$

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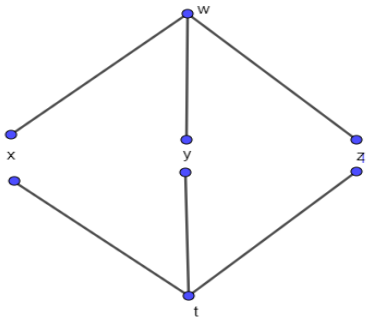


➢ $\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$

$= (y^2)(-sint) + (2xy)(cost) + (2z)(1)$

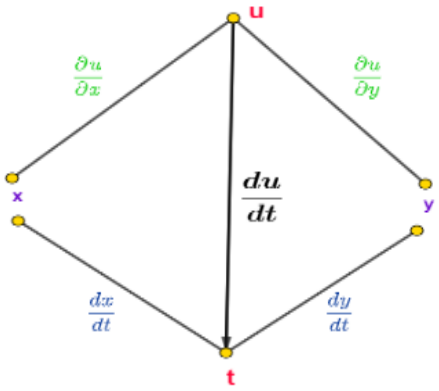
$= (sin^2 t)(-sint) + (2sint.cost)(cost) + (2t)(1)$

$= -sin^3 t + 2sint.cos^2 t + 2t$

$\therefore \left(\frac{dw}{dt}\right)_{t=0} = 0 + 0 + 0 = 0$

Example 5: If $u = \tan^{-1}\left(\frac{y}{x}\right), x = e^t - e^{-t}, y = e^t + e^{-t}$ then find $\frac{du}{dt}$

Solution: $u \to x, y \to t$



➢ $\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$

$= \frac{1}{1+\frac{y^2}{x^2}} \cdot \left(-\frac{y}{x^2}\right) \cdot (e^t + e^{-t}) + \frac{1}{1+\frac{y^2}{x^2}} \cdot \left(\frac{1}{x}\right) \cdot (e^t - e^{-t})$

$= -\frac{y}{x^2+y^2} \cdot y + \frac{x}{x^2+y^2} \cdot x$

$= \frac{x^2-y^2}{x^2+y^2}$

$= \frac{(e^t-e^{-t})^2-(e^t+e^{-t})^2}{(e^t-e^{-t})^2+(e^t+e^{-t})^2}$

$$= \frac{e^{2t}-2+e^{-2t}-(e^{2t}+2+e^{-2t})}{e^{2t}-2+e^{-2t}+e^{2t}+2+e^{-2t}}$$

$$= -\frac{4}{2e^{2t}+2e^{-2t}}$$

$$\frac{du}{dt} = -\frac{2}{e^{2t}+e^{-2t}}$$

H.W.

1) If $u = x^2 + y^2;\ x = at^2, y = 2at$ then find $\frac{du}{dt}$

   Ans: $4a^2t(t^2+2)$

2) If $u = x^2 + y^2 + z^2;\ x = e^t, y = e^t sint, z = e^t cost$ then find $\frac{du}{dt}$

   Ans:$4e^{2t}$

3) If $z = e^{xy}$ $where\ x = tcost\ , y = tsint$ then find $\frac{dz}{dt}$ $at\ t = \frac{\pi}{2}$

   Ans:$-\frac{\pi^2}{4}$

4) If $u = sin\left(\frac{x}{y}\right); x = e^t, y = t^2$ then find $\frac{du}{dt}$

   Ans:$\frac{1}{t^2}e^t cos\left(\frac{e^t}{t^2}\right)\left[1-\frac{2}{t}\right]$

5) If $u = \sin^{-1}(x-y)\ ; x = 3t\ , y = 4t^3$ Prove that $\frac{dz}{dt} = \frac{3}{\sqrt{1-t^2}}$.

6) If $u = sin^{-1}(x-y), x = 3t$ and $y = 4t$ then find $\frac{du}{dt}$.

   Ans: $\frac{-1}{\sqrt{1-t^2}}$

7) If $z = xy^2 + x^2y\ , where\ x = at^2\ \&\ y = 2at$ then find $\frac{dz}{dt}$.

   Ans:$2a^3t^3(8+5t)$

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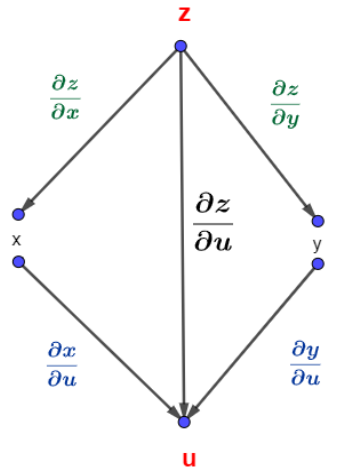
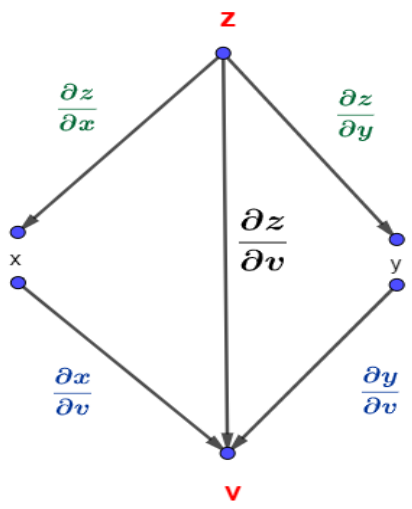
❖ Composite function of two variable:

- Let $z = f(x, y)$ and $x = \phi(u, v), y = \Psi(u, v)$.

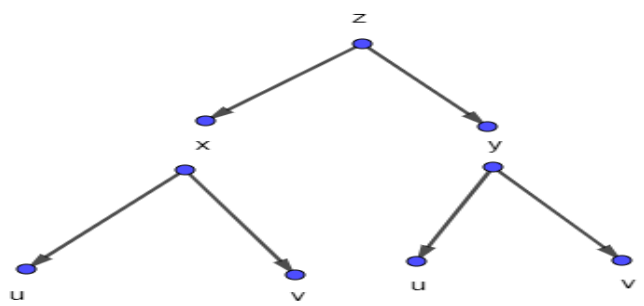
  $z \to x, y \to u, v$
- Here, z is a function of x, y and $x, y$are functions of u and v, so these relations define z as a function of u, v. Thus, z is called as a composite function of u and v.
- z has partial derivatives $w.r.t\ \ u$ & v given by using chain rule,



$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}.\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}.\frac{\partial y}{\partial u}$$



$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}.\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}.\frac{\partial y}{\partial v}$$



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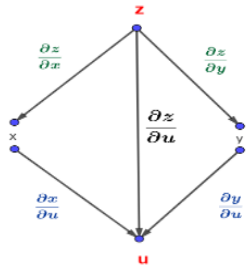
Example1: If $z = f(x, y)$ ***and*** $x = e^{u} + e^{-v}, y = e^{-u} - e^{v}$ **,Show that**

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$$

Solution: $z \to x, y \to u, v$

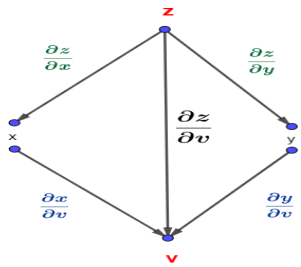
- $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} = \frac{\partial z}{\partial x}(e^{u}) + \frac{\partial z}{\partial y}(-e^{-u})$

$$---(1)$$



- $\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v} = \frac{\partial z}{\partial x}(-e^{-v}) + \frac{\partial z}{\partial y}(-e^{v})$

$$---(2)$$



- From (1) & (2),

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = \left[\frac{\partial z}{\partial x}(e^{u}) + \frac{\partial z}{\partial y}(-e^{-u})\right] - \left[\frac{\partial z}{\partial x}(-e^{-v}) + \frac{\partial z}{\partial y}(-e^{v})\right]$$

$$= \frac{\partial z}{\partial x}(e^{u}) + \frac{\partial z}{\partial y}(-e^{-u}) + \frac{\partial z}{\partial x}(e^{-v}) + \frac{\partial z}{\partial y}(e^{v})$$

$$= \frac{\partial z}{\partial x}(e^{u} + e^{-v}) + \frac{\partial z}{\partial y}(e^{-u} - e^{-v})$$

$$= x.\frac{\partial z}{\partial x} - y.\frac{\partial z}{\partial y}$$

Example 2: If $z = tan^{-1}\left(\frac{x}{y}\right), x = ucosv, y = usinv$ **evaluate** $\frac{\partial z}{\partial u}$ **and** $\frac{\partial z}{\partial v}$

Solution: $z \to x, y \to u, v$

- $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}.\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}.\frac{\partial y}{\partial u}$

$$= \left[\frac{\frac{1}{y}}{1+\left(\frac{x}{y}\right)^{2}}\right](cosv) + \left[\frac{\left(-\frac{x}{y^{2}}\right)}{1+\left(\frac{x}{y}\right)^{2}}\right](sinv)$$

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$$= \left[\frac{y}{x^2+y^2}\right](cosv) + \left[\frac{-x}{x^2+y^2}\right](sinv)$$

$$= \left[\frac{usinv}{(ucosv)^2+(usinv)^2}\right](cosv) + \left[\frac{-ucosv}{(ucosv)^2+(usinv)^2}\right](sinv)$$

$$= \frac{usinv.cosv}{u^2} - \frac{usinv.cosv}{u^2}$$

$$= 0$$

- $\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}.\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}.\frac{\partial y}{\partial v}$

$$= \left[\frac{\frac{1}{y}}{1+\left(\frac{x}{y}\right)^2}\right](-usinv) + \left[\frac{\left(-\frac{x}{y^2}\right)}{1+\left(\frac{x}{y}\right)^2}\right](ucosv)$$

$$= \left[\frac{y}{x^2+y^2}\right](-usinv) + \left[\frac{-x}{x^2+y^2}\right](ucosv)$$

$$= \left[\frac{usinv}{u^2}\right](-usinv) + \left[\frac{-ucosv}{u^2}\right](ucosv)$$

$$= -(sin^2v + cos^2v)$$

$$= -1$$

Example 3: Express $\frac{\partial w}{\partial r}$ & $\frac{\partial w}{\partial s}$ in terms of r&s If $w = x + 2y + z^2$;

$$x = \frac{r}{s}, y = r^2 + lns, z = 2r.$$

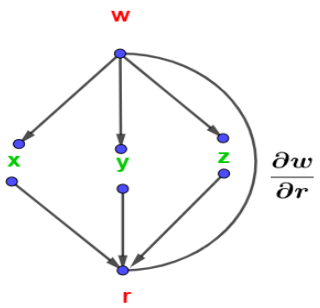
Solution: $w \to x, y, z \to r, s$

- $\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}.\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}.\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}.\frac{\partial z}{\partial r}$

$$= (1)\left(\frac{1}{s}\right) + (2)(2r) + (2z)(2)$$

$$= \frac{1}{s} + 4r + 4(2r)$$

$$= \frac{1}{s} + 12r$$

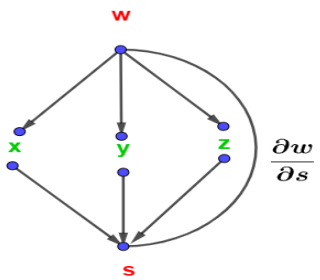


- $\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}.\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}.\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}.\frac{\partial z}{\partial s}$

$$= (1)\left(\frac{-r}{s^2}\right) + (2)\left(\frac{1}{s}\right) + (2z)(0)$$

$$= -\frac{r}{s^2} + \frac{2}{s}$$

$$= \frac{1}{s}\left(2 - \frac{r}{s}\right)$$



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Example 4: If $w = x^2 + y^2, x = r - s, y = r + s$, using chain rule ,

prove that $\frac{\partial w}{\partial s} = 4s$.

Solution: $w \to x, y \to r, s$

- $\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}.\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}.\frac{\partial y}{\partial s}$

$$= (2x)(-1) + (2y)(1)$$
$$= -2(r - s) + 2(r + s)$$
$$= -2r + 2s + 2r + 2s$$
$$= 4s$$

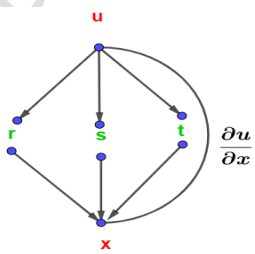
Example 5: If $u = f(x - y, y - z, z - x)$ Show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Solution:

- Let $r = x - y\,, s = y - z, t = z - x$
- $u = f(r, s, t)$
- $u \to r, s, t \to x, y, z$

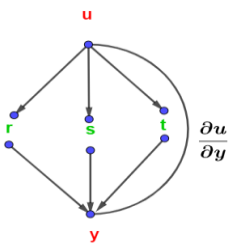
- $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial x}$

$$= \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(0) + \frac{\partial u}{\partial t}(-1)$$
$$= \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \qquad \ldots\ldots\ldots(1)$$



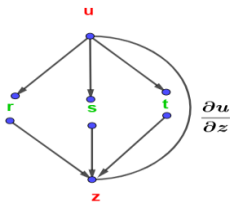
- $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial y}$

$$= \frac{\partial u}{\partial r}(-1) + \frac{\partial u}{\partial s}(1) + \frac{\partial u}{\partial t}(0)$$
$$= -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \qquad \ldots\ldots\ldots (2)$$



- $\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial z} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial z}$

$$= \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}(-1) + \frac{\partial u}{\partial t}(1)$$
$$= -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \qquad \ldots\ldots\ldots (3)$$



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Adding (1), (2) & (3),

$$\Rightarrow \frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{\partial u}{\partial r}-\frac{\partial u}{\partial t}-\frac{\partial u}{\partial r}+\frac{\partial u}{\partial s}-\frac{\partial u}{\partial s}+\frac{\partial u}{\partial t}=0$$

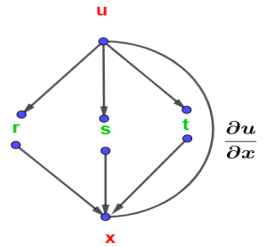
Example 6: If $u=f\left(\frac{x}{y},\frac{y}{z},\frac{z}{x}\right)$ Show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=0$

Solution:

- Let $r=\frac{x}{y}, s=\frac{y}{z}, t=\frac{z}{x}$
- $u=f(r,s,t)$
- $u\rightarrow r,s,t\rightarrow x,y,z$
- $\frac{\partial u}{\partial x}=\frac{\partial u}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial u}{\partial s}\frac{\partial s}{\partial x}+\frac{\partial u}{\partial t}\frac{\partial t}{\partial x}$

  $=\frac{\partial u}{\partial r}\left(\frac{1}{y}\right)+\frac{\partial u}{\partial s}(0)+\frac{\partial u}{\partial t}\left(-\frac{z}{x^2}\right)$

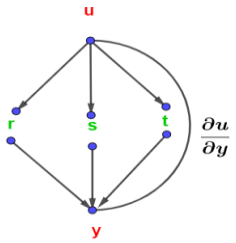
  $\therefore x.\frac{\partial u}{\partial x}=\frac{x}{y}\frac{\partial u}{\partial r}-\frac{z}{x}\frac{\partial u}{\partial t}$ .....(1)



- $\frac{\partial u}{\partial y}=\frac{\partial u}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial u}{\partial s}\frac{\partial s}{\partial y}+\frac{\partial u}{\partial t}\frac{\partial t}{\partial y}$

  $=\frac{\partial u}{\partial r}\left(-\frac{x}{y^2}\right)+\frac{\partial u}{\partial s}\left(\frac{1}{z}\right)+\frac{\partial u}{\partial t}(0)$

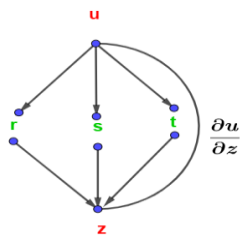
  $\therefore y.\frac{\partial u}{\partial y}=-\frac{x}{y}\frac{\partial u}{\partial r}+\frac{y}{z}\frac{\partial u}{\partial s}$ ..... (2)



- $\frac{\partial u}{\partial z}=\frac{\partial u}{\partial r}\frac{\partial r}{\partial z}+\frac{\partial u}{\partial s}\frac{\partial s}{\partial z}+\frac{\partial u}{\partial t}\frac{\partial t}{\partial z}$

  $=\frac{\partial u}{\partial r}(0)+\frac{\partial u}{\partial s}\left(-\frac{y}{z^2}\right)+\frac{\partial u}{\partial t}\left(\frac{1}{x}\right)$

  $\therefore z.\frac{\partial u}{\partial z}=-\frac{y}{z}\frac{\partial u}{\partial s}+\frac{z}{x}\frac{\partial u}{\partial t}$ ..... (3)



Adding (1), (2) & (3),

$$\Rightarrow x.\frac{\partial u}{\partial x}+y.\frac{\partial u}{\partial y}+z.\frac{\partial u}{\partial z}=\frac{x}{y}\frac{\partial u}{\partial r}-\frac{z}{x}\frac{\partial u}{\partial t}-\frac{x}{y}\frac{\partial u}{\partial r}+\frac{y}{z}\frac{\partial u}{\partial s}-\frac{y}{z}\frac{\partial u}{\partial s}+\frac{z}{x}\frac{\partial u}{\partial t}=0$$

Example 7: If $u=f\left(\frac{y-x}{xy},\frac{z-x}{xz}\right)$, show that $x^2\frac{\partial u}{\partial x}+y^2\frac{\partial u}{\partial y}+z^2\frac{\partial u}{\partial z}=0$

Solution:

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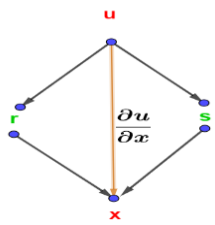
- Let $r = \frac{y-x}{xy} = \frac{1}{x} - \frac{1}{y}, s = \frac{z-x}{xz} = \frac{1}{x} - \frac{1}{z}$
- $u = f(r, s)$
- $u \to r, s \to x, y, z$

- $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x}$

  $= \frac{\partial u}{\partial r}\left(-\frac{1}{x^2}\right) + \frac{\partial u}{\partial s}\left(-\frac{1}{x^2}\right)$

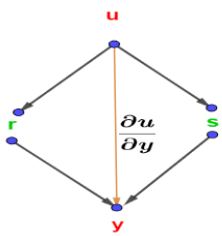
  $\therefore x^2\frac{\partial u}{\partial x} = -\frac{\partial u}{\partial r} - \frac{\partial u}{\partial s}$ .....(1)



- $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y}$

  $= \frac{\partial u}{\partial r}\left(\frac{1}{y^2}\right) + \frac{\partial u}{\partial s}(1) + \frac{\partial u}{\partial t}(0)$

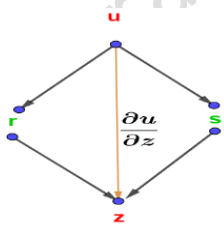
  $\therefore y^2\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}$ ..... (2)



- $\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial z}$

  $= \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}\left(\frac{1}{z^2}\right)$

  $\therefore z^2\frac{\partial u}{\partial z} = \frac{\partial u}{\partial s}$ ..... (3)



Adding (1), (2) & (3),

$$\Rightarrow x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial r} - \frac{\partial u}{\partial s} + \frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} = 0$$

H.W.

1) If $z = f(x, y)$ *and* $x = uv,\ y = \frac{u+v}{u-v}$ Show that valuate $u\frac{\partial z}{\partial u} + v\frac{\partial z}{\partial v} = 2x\frac{\partial z}{\partial x}$
2) If $u = f(x^2 - y^2, y^2 - z^2, z^2 - x^2)$ then prove that $\frac{1}{x}\frac{\partial u}{\partial x} + \frac{1}{y}\frac{\partial u}{\partial y} + \frac{1}{z}\frac{\partial u}{\partial z} = 0$
3) If $u = f(e^{y-z}, e^{z-x}, e^{x-y})$ Show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$
4) Express $\frac{\partial w}{\partial r}$ & $\frac{\partial w}{\partial s}$ in terms of r & s, If $w = x^2 + y^2$; $x = r - s$, $y = r + s$.
5) If $u = f(x^2 + 2yz, y^2 + 2zx)$ then prove that

   $$(y^2 - zx)\frac{\partial u}{\partial x} + (x^2 - yz)\frac{\partial u}{\partial y} + (z^2 - xy)\frac{\partial u}{\partial z} = 0$$

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❖ Jacobian:

If u & v are functions of two independent variables x & y then the determinant

$$J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

Is called Jacobian of u & v $w.r.t$ x & y.

Similarly,

u , v & w are functions of three independent variables x , y & z then the determinant

$$J = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial y} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Is called Jacobian of u , v & w $w.r.t$ x , y & z.

Example 1: Find Jacobian $\frac{\partial(x,y)}{\partial(r,\theta)}$ for $x = r\cos\theta$, $y = r\sin\theta$

Solution:

- $J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$

$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$

$= r\cos^2\theta + r\sin^2\theta$

$= r(\cos^2\theta + r\sin^2\theta)$

$= r$

Example 2: Find Jacobian $\frac{\partial(x,y,z)}{\partial(r,\theta,\emptyset)}$ for $x = r\sin\theta\cos\emptyset, y = r\sin\theta\sin\emptyset, z = r\cos\theta$

Solution:

$$J = \frac{\partial(x,y,z)}{\partial(r,\theta,\emptyset)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \emptyset} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \emptyset} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial z}{\partial \emptyset} \end{vmatrix} = \begin{vmatrix} \sin\theta\cos\emptyset & r\cos\theta\cos\emptyset & -r\sin\theta\sin\emptyset \\ \sin\theta\sin\emptyset & r\cos\theta\sin\emptyset & r\sin\theta\cos\emptyset \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix}$$

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$$= sin\theta cos\emptyset(0 + r^2 sin^2\theta cos\emptyset) - rcos\theta cos\emptyset(0 - rsin\theta cos\theta cos\emptyset)$$
$$- rsin\theta sin\emptyset(-rsin^2\theta sin\emptyset - rcos^2\theta sin\emptyset)$$

$$= r^2 sin^3\theta cos^2\emptyset + r^2 sin\theta cos^2\theta cos^2\emptyset + r^2 sin^3\theta sin^2\emptyset + r^2 sin\theta cos^2\theta sin^2\emptyset$$

$$= r^2 sin^3\theta(cos^2\emptyset + sin^2\emptyset) + r^2 sin\theta cos^2\theta(cos^2\emptyset + sin^2\emptyset)$$

$$= r^2 sin^3\theta + r^2 sin\theta cos^2\theta$$

$$= r^2 sin\theta(sin^2\theta + cos^2\theta)$$

$$= r^2 sin\theta$$

❖ Tangent plane & Normal line:

- Let the equation of the surface be $f(x, y, z) = 0$
- The equation of tangent plane to the surface $f(x, y, z) = 0$ at point $P(x_0, y_0, z_0)$ is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

- The equation of Normal line to the surface $f(x, y, z) = 0$ at point $P(x_0, y_0, z_0)$ is,

$$\frac{(x - x_0)}{\left(\frac{\partial f}{\partial x}\right)_p} = \frac{(y - y_0)}{\left(\frac{\partial f}{\partial y}\right)_p} = \frac{(z - z_0)}{\left(\frac{\partial f}{\partial z}\right)_p}$$

Example 1: Find the equation of tangent plane and normal line to the surface $x^2 + y^2 + z^2 = 3$ at the point $(1, 1, 1)$.

Solution:

- let $f(x, y, z) = x^2 + y^2 + z^2 - 3 = 0$

$\frac{\partial f}{\partial x} = 2x$ $\qquad \left(\frac{\partial f}{\partial x}\right)_p = 2$

$\frac{\partial f}{\partial y} = 2y$ $\qquad \left(\frac{\partial f}{\partial y}\right)_p = 2$

$\frac{\partial f}{\partial z} = 2z$ $\qquad \left(\frac{\partial f}{\partial z}\right)_p = 2$

- The equation of tangent plane is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

$$(x - 1)2 + (y - 1)2 + (z - 1)2 = 0$$

$$2x - 2 + 2y - 2 + 2z - 2 = 0$$

$$2x + 2y + 2z - 6 = 0$$

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$\therefore x + y + z = 3$

- The equation of normal line is,

$$\frac{(x - x_0)}{\left(\frac{\partial f}{\partial x}\right)_p} = \frac{(y - y_0)}{\left(\frac{\partial f}{\partial y}\right)_p} = \frac{(z - z_0)}{\left(\frac{\partial f}{\partial z}\right)_p}$$

$$\frac{(x - 1)}{2} = \frac{(y - 1)}{2} = \frac{(z - 1)}{2}$$

$$\therefore x - 1 = y - 1 = z - 1$$

Example 2: Find Tangent plane and normal line of $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$ at the point $(-2, 1, -3)$

Solution:

- let $f(x, y, z) = \frac{x^2}{4} + y^2 + \frac{z^2}{9} - 3 = 0$

$\frac{\partial f}{\partial x} = \frac{2x}{4} = \frac{x}{2}$ $\qquad \left(\frac{\partial f}{\partial x}\right)_p = -\frac{2}{2} = -1$

$\frac{\partial f}{\partial y} = 2y$ $\qquad \left(\frac{\partial f}{\partial y}\right)_p = 2(1) = 2$

$\frac{\partial f}{\partial z} = \frac{2z}{9}$ $\qquad \left(\frac{\partial f}{\partial z}\right)_p = \frac{2(-3)}{9} = -\frac{2}{3}$

- The equation of tangent plane is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

$$(x + 2)(-1) + (y - 1)(2) + (z + 3)\left(-\frac{2}{3}\right) = 0$$

$$-x - 2 + 2y - 2 + \left(\frac{-2z-6}{3}\right) = 0$$

$$-3x + 6y - 2z - 18 = 0$$

$$\Rightarrow 3x - 6y + 2z = -18$$

- The equation of normal line is,

$$\frac{(x - x_0)}{\left(\frac{\partial f}{\partial x}\right)_p} = \frac{(y - y_0)}{\left(\frac{\partial f}{\partial y}\right)_p} = \frac{(z - z_0)}{\left(\frac{\partial f}{\partial z}\right)_p}$$

$$\therefore \frac{(x + 2)}{-1} = \frac{(y - 1)}{2} = \frac{(z + 3)}{\left(-\frac{2}{3}\right)}$$

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Example 3: Find the equation of the tangent plane and normal line to the surface $2xz^2 - 3xy - 4x = 7$ *at* the *point* $(1, -1, 2)$

- let $f(x, y, z) = 2xz^2 - 3xy - 4x - 7 = 0$

$\frac{\partial f}{\partial x} = 2z^2 - 3y - 4$ $\qquad \left(\frac{\partial f}{\partial x}\right)_p = 7$

$\frac{\partial f}{\partial y} = -3x$ $\qquad \left(\frac{\partial f}{\partial y}\right)_p = -3$

$\frac{\partial f}{\partial z} = 4xz$ $\qquad \left(\frac{\partial f}{\partial z}\right)_p = 8$

- The equation of tangent plane is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

$(x - 1)7 + (y + 1)(-3) + (z - 2)8 = 0$

$7x - 7 - 3y - 3 + 8z - 16 = 0$

$7x - 3y + 8z - 26 = 0$

$\therefore 7x - 3y + 8z = 26$

- The equation of normal line is,

$$\frac{(x - x_0)}{\left(\frac{\partial f}{\partial x}\right)_p} = \frac{(y - y_0)}{\left(\frac{\partial f}{\partial y}\right)_p} = \frac{(z - z_0)}{\left(\frac{\partial f}{\partial z}\right)_p}$$

$$\frac{(x - 1)}{7} = \frac{(y + 1)}{-3} = \frac{(z - 2)}{8}$$

Example 4: Find the tangent plane and normal line to the surface

$x^2yz + 3y^2 = 2xz^2 - 8z$ at the point $p(1, 2, -1)$.

- **Solution:** let $f(x, y, z) = x^2yz + 3y^2 - 2xz^2 + 8z = 0$

$\frac{\partial f}{\partial x} = 2xyz - 2z^2$ $\qquad \left(\frac{\partial f}{\partial x}\right)_p = 2(1)(2)(-1) - 2(-1)^2 = -6$

$\frac{\partial f}{\partial y} = x^2z + 6y$ $\qquad \left(\frac{\partial f}{\partial y}\right)_p = 1(-1) + 6(2) = 11$

$\frac{\partial f}{\partial z} = x^2y - 4xz + 8$ $\qquad \left(\frac{\partial f}{\partial z}\right)_p = 1(2) - 4(1)(-1) + 8 = 14$

- The equation of tangent plane is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

$(x - 1)(-6) + (y - 2)(11) + (z + 1)(14) = 0$

$-6x + 6 + 11y - 22 + 14z + 14 = 0$

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$-6x + 11y + 14z - 2 = 0$

$\therefore -6x + 11y + 14z = 2 \Rightarrow 6x - 11y - 14z = -2$

- The equation of normal line is,

$$\frac{(x - x_0)}{\left(\frac{\partial f}{\partial x}\right)_p} = \frac{(y - y_0)}{\left(\frac{\partial f}{\partial y}\right)_p} = \frac{(z - z_0)}{\left(\frac{\partial f}{\partial z}\right)_p}$$

$$\frac{(x - 1)}{-6} = \frac{(y - 2)}{11} = \frac{z + 1}{14}$$

Example 5: Find the equations of tangent plane and normal line to surface

$\boldsymbol{\frac{x^2}{2} - \frac{y^2}{3} - z = 0 \; at \; (2, 3, -1).}$

Solution: let $f(x, y, z) = \frac{x^2}{2} - \frac{y^2}{3} - z = 0$

$\frac{\partial f}{\partial x} = \frac{2x}{2} = x$ $\qquad \left(\frac{\partial f}{\partial x}\right)_p = 2$

$\frac{\partial f}{\partial y} = \frac{-2y}{3}$ $\qquad \left(\frac{\partial f}{\partial y}\right)_p = -\frac{2(3)}{3} = -2$

$\frac{\partial f}{\partial z} = -1$ $\qquad \left(\frac{\partial f}{\partial z}\right)_p = -1$

- The equation of tangent plane is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

$(x - 2)(2) + (y - 3)(-2) + (z + 1)(-1) = 0$

$2x - 4 - 2y + 6 - z - 1 = 0$

$2x - 2y - z + 1 = 0$

$\therefore 2x - 2y - z = -1$

- The equation of normal line is,

$$\frac{(x - x_0)}{\left(\frac{\partial f}{\partial x}\right)_p} = \frac{(y - y_0)}{\left(\frac{\partial f}{\partial y}\right)_p} = \frac{(z - z_0)}{\left(\frac{\partial f}{\partial z}\right)_p}$$

$$\frac{(x - 2)}{2} = \frac{(y - 3)}{-2} = \frac{z + 1}{-1}$$

Example 6: Find the tangent plane and normal line to the surface

$\boldsymbol{z = 3x^2 - xy}$ **at the point** $\boldsymbol{(1, 2, 1)}$.

Solution: let $f(x, y, z) = 3x^2 - xy - z = 0$

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$\frac{\partial f}{\partial x} = 6x - y$ $\left(\frac{\partial f}{\partial x}\right)_p = 6(1) - 2 = 4$

$\frac{\partial f}{\partial y} = -x$ $\left(\frac{\partial f}{\partial y}\right)_p = -1$

$\frac{\partial f}{\partial z} = -1$ $\left(\frac{\partial f}{\partial z}\right)_p = -1$

- The equation of tangent plane is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

$(x - 1)(4) + (y - 2)(-1) + (z - 1)(-1) = 0$

$4x - 4 - y + 2 - z + 1 = 0$

$4\mathrm{x} - \mathrm{y} - \mathrm{z} - 1 = 0$

$\therefore 4\mathrm{x} - \mathrm{y} - \mathrm{z} = 1$

- The equation of normal line is,

$$\frac{(x - x_0)}{\left(\frac{\partial f}{\partial x}\right)_p} = \frac{(y - y_0)}{\left(\frac{\partial f}{\partial y}\right)_p} = \frac{(z - z_0)}{\left(\frac{\partial f}{\partial z}\right)_p}$$

$$\frac{(x - 1)}{4} = \frac{(y - 2)}{-1} = \frac{z - 1}{-1}$$

Example 7: Show that $z = xy - 2$ *and* $x^2 + y^2 + z^2 = 3$ have same tangent plane at the point $(1, 1, -1)$

Solution:

- let $f(x, y, z) = xy - 2 - z = 0$

$\frac{\partial f}{\partial x} = y$ $\left(\frac{\partial f}{\partial x}\right)_p = 1$

$\frac{\partial f}{\partial y} = x$ $\left(\frac{\partial f}{\partial y}\right)_p = 1$

$\frac{\partial f}{\partial z} = -1$ $\left(\frac{\partial f}{\partial z}\right)_p = -1$

- The equation of tangent plane is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

$(x - 1)1 + (y - 1)1 + (z + 1)(-1) = 0$

$x - 1 + y - 1 - z - 1 = 0$

$x + y - z - 3 = 0$

$\therefore x + y - z = 3 \quad \ldots\ldots\ldots (1)$

- let $f(x, y, z) = x^2 + y^2 + z^2 - 3 = 0$

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$\frac{\partial f}{\partial x} = 2x$ $\left(\frac{\partial f}{\partial x}\right)_p = 2$

$\frac{\partial f}{\partial y} = 2y$ $\left(\frac{\partial f}{\partial y}\right)_p = 2$

$\frac{\partial f}{\partial z} = 2z$ $\left(\frac{\partial f}{\partial z}\right)_p = -2$

- The equation of tangent plane is,

$$(x - x_0)\left(\frac{\partial f}{\partial x}\right)_p + (y - y_0)\left(\frac{\partial f}{\partial y}\right)_p + (z - z_0)\left(\frac{\partial f}{\partial z}\right)_p = 0$$

$(x - 1)2 + (y - 1)2 + (z + 1)(-2) = 0$

$2x - 2 + 2y - 2 - 2z - 2 = 0$

$2x + 2y - 2z - 6 = 0$

$\therefore x + y - z = 3 \quad \ldots\ldots\ldots (2)$

- From (1) and (2),
- $z = xy - 2$ *and* $x^2 + y^2 + z^2 = 3$ have same tangent plane at the point $(1,1,-1)$.

H.W.

1) **Find Tangent plane and normal line of $xyz = 6$ at the point $(1, 2, 3)$**

   Ans: $6x + 3y + 2z = 18$ & $\frac{(x-1)}{6} = \frac{(y-2)}{3} = \frac{(z-3)}{2}$

2) **Find Tangent plane and normal line of $cos\pi x - x^2y + e^{xz} + yz = 4$at the point $(0, 1, 2)$**

   Ans: $2x + 2y + z = -2$ & $\frac{(x-0)}{2} = \frac{(y-1)}{2} = \frac{(z-2)}{1}$

3) **Find Tangent plane and normal line of $2z - x^2 = 0$the point $(2, 0, 2)$**

   Ans: $2x - z = 2$ & $\frac{(x-2)}{-4} = \frac{(y-0)}{0} = \frac{(z-2)}{2}$

4) **Find Tangent plane and normal line of $2x^2 + y^2 + 2z - 3 = 0$ at the point $(2, 1, -3)$**

   Ans: $4x + y + z = 0$ & $\frac{(x-2)}{8} = \frac{(y-1)}{2} = \frac{z+3}{2}$

5) **Find Tangent plane and normal line of $x^2 + y^2 + z^2 - 9 = 0$ at the point $(1, 2, 4)$**

   Ans: $2x + 4y + z = 14$ & $\frac{(x-1)}{2} = \frac{(y-2)}{4} = \frac{z-4}{1}$

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❖ Local Extreme values (Maxima & Minima):

- A function $f(x,y)$ is said to be **maximum** at (a,b) if $f(a,b) \geq f(x,y); \forall (x,y) \in D$.
- A function $f(x,y)$ is said to be **minimum** at (a,b) if $f(a,b) \leq f(x,y); \forall (x,y) \in D$.
- Maximum or Minimum values of a function is called its **Extreme values.**
- The point at which both f_x & f_y are zero are called **stationary points** or **Critical points** or **turning points**.

  $i.e$ if $f_x(a,b) = 0$ & $f_y(a,b) = 0$ then (a,b) is called stionary points.

- **Working rule to find Maximum & minimum values of a $f(x,y)$:**

To determine maximum & minimum values of the function $f(x,y)$

Step-1: solve $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial x} = 0$ simultaneously for x & y This gives the stationary points (a_1,b_1) , (a_2,b_2) , (a_3,b_3), … …

Step-2: $= \frac{\partial^2 f}{\partial x^2} = f_{xx}$, $s = \frac{\partial^2 f}{\partial x \partial y} = f_{xy}$, $t = \frac{\partial^2 f}{\partial y^2} = f_{yy}$

Obtain values of r, s, t at stationary points (a_1,b_1) , (a_2,b_2) , (a_3,b_3), … …

Step-3: Find $rt - s^2$ at all stationary points.

i. If $rt - s^2 > 0$ & $r < 0$ at (a,b) then $f(x,y)$ is maximum at (a,b) & maximum value of $f(x,y)$ is $f(a,b)$.

ii. It $rt - s^2 > 0$ & $r > 0$ at (a,b) then $f(x,y)$ is minimum at (a,b) & minimum value of $f(x,y)$ is $f(a,b)$.

iii. $rt - s^2 < 0$ at (a,b) then $f(x,y)$ is neither maximum nor minimum at (a,b). This point is known as **saddle point.**

iv. $rt - s^2 = 0$ at (a,b) then no conclusion can be made about the extreme values of $f(x,y)$.

- Every extreme value is stationary value but converse may not be true.

		$f(x,y)$	
$rt - s^2 > 0$	$r < 0$	maximum	Extreme value
$rt - s^2 > 0$	$r > 0$	minimum	Extreme value
$rt - s^2 < 0$	–	Neither maximum nor minimum	Saddle point
$rt - s^2 = 0$	–	No conclusion	–

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Example 1: Find extreme value of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

Solution: $f(x,y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$

- **Step 1:**

$$\Rightarrow \frac{\partial f}{\partial x} = 0$$

$$3x^2 + 3y^2 - 30x + 72 = 0$$

$$3(x^2 + y^2 - 10x + 24) = 0$$

$$(x^2 + y^2 - 10x + 24) = 0 \ldots\ldots\ldots (1)$$

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

$$6xy - 30y = 0$$

$$6y(x-5) = 0$$

$$y = 0 \text{ or } x = 5 \ldots\ldots\ldots (2)$$

a) **When $y = 0$** $\Rightarrow$ *From Eq*$(1) \Rightarrow x^2 + 0 - 10x + 24 = 0$

$$\Rightarrow (x-4)(x-6) = 0$$

$$\Rightarrow x = 4,6$$

b) **When $x = 5$** $\Rightarrow$ *From Eq*$(1) \Rightarrow 25 + y^2 - 50 + 24 = 0$

$$\Rightarrow y^2 - 1 = 0$$

$$\Rightarrow y = 1, -1$$

$\therefore$ ***Stationary points*** : $(4,0), (6,0), (5,1), (5,-1)$

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 6x - 30$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 6y$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6x - 30$$

- **Step 3:**

(x,y)	r	s	t	$rt - s^2$	Conclusion	f at (a,b)
(4,0)	$-6 < 0$	0	-6	$36 > 0$	Maximum	112
(6,0)	$6 > 0$	0	6	$36 > 0$	Minimum	108
(5,1)	0	6	0	$-36 < 0$	Saddle point	---
$(5,-1)$	0	-6	0	$-36 < 0$	Saddle point	---

$\Rightarrow f_{max} = f_{(4,0)} = (4)^3 + 0 - 15(4)^2 + 72(4) = 64 - 240 + 288 = 112$

$\Rightarrow f_{min} = f_{(6,0)} = (6)^3 + 0 - 15(6)^2 + 72(6) = 216 - 540 + 432 = 108$

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Example 2: Find extreme value $x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$

Solution: $f(x,y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$

Step 1:

$$\Rightarrow \frac{\partial f}{\partial x} = 0$$

$$3x^2 + 3y^2 - 6x = 0$$

$$3(x^2 + y^2 - 2x) = 0$$

$$(x^2 + y^2 - 2x) = 0 \ldots\ldots\ldots (1)$$

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

$$6xy - 6y = 0$$

$$6y(x-1) = 0$$

$$y(x-1) = 0$$

$$y = 0 \text{ or } x = 1 \ldots\ldots\ldots (2)$$

a) **When $y = 0$** $\Rightarrow$ *From* $Eq(1) \Rightarrow x^2 + 0 - 2x = 0$

$$\Rightarrow x(x-2) = 0$$

$$\Rightarrow x = 0,2$$

b) **When $x = 1$** $\Rightarrow$ *From* $Eq(1) \Rightarrow 1 + y^2 - 2 = 0$

$$\Rightarrow y^2 - 1 = 0$$

$$\Rightarrow y = 1, -1$$

$\therefore$ ***Stationary points*** : $(0,0), (2,0), (1,1), (1,-1)$

➢ **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 6x - 6$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 6y$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6x - 6$$

➢ **Step 3:**

(x,y)	r	s	t	$rt - s^2$	Conclusion	f at (a,b)
(0,0)	$-6 < 0$	0	-6	$36 > 0$	Maximum	4
(2,0)	$6 > 0$	0	6	$36 > 0$	Minimum	0
(1,1)	0	6	0	$-36 < 0$	Saddle point	---
$(1,-1)$	0	-6	0	$-36 < 0$	Saddle point	---

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$\Longrightarrow f_{max} = f_{(0,0)} = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4 = (0)^3 + 0 - 0 - 0 + 4 = 4$

$\Longrightarrow f_{min} = f_{(2,0)} = (2)^3 + 0 - 3(2)^2 - 0 + 4 = 8 - 12 + 4 = 0$

Example 3: Find all the points of maxima & minima of $f(x,y) = x + y + \frac{1}{x} + \frac{1}{y}$, Also find the maximum & minimum values of the function.

Solution: $f(x,y) = x + y + \frac{1}{x} + \frac{1}{y}$

Step 1:

$$\Longrightarrow \frac{\partial f}{\partial x} = 0$$

$$1 - \frac{1}{x^2} = 0$$

$$\frac{x^2-1}{x^2} = 0$$

$$x^2 - 1 = 0 \ldots\ldots\ldots (1)$$

$$\Longrightarrow x = \pm 1$$

$$\Longrightarrow \frac{\partial f}{\partial y} = 0$$

$$1 - \frac{1}{y^2} = 0$$

$$\frac{y^2-1}{y^2} = 0$$

$$y^2 - 1 = 0 \ldots\ldots\ldots (1)$$

$$\Longrightarrow y = \pm 1$$

$\therefore$ ***Stationary points*** : $(1,1), (-1,-1), (1,-1), (-1,1)$

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = \frac{2}{x^3}$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$t = \frac{\partial^2 f}{\partial y^2} = \frac{2}{x^3}$$

- **Step 3:**

(x,y)	r	s	t	$rt - s^2$	Conclusion	f at (a,b)
$(1,1)$	$2 > 0$	0	2	$4 > 0$	Minimum	4
$(-1,-1)$	$-2 < 0$	0	-2	$4 > 0$	Maximum	-4
$(1,-1)$	2	0	-2	$-4 < 0$	Saddle point	---
$(-1,1)$	-2	0	2	$-4 < 0$	Saddle point	---

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$\Longrightarrow f_{min} = f_{(1,1)} = 1 + 1 + 1 + 1 = 4$

$\Longrightarrow f_{max} = f_{(-1,-1)} = -1 - 1 - 1 - 1 = -4$

Example 4: Find the extreme values of the function

$\boldsymbol{f(x,y) = x^3 + y^3 - 3x - 12y + 20}$

Solution: $f(x,y) = x^3 + y^3 - 3x - 12y + 20$

Step 1:

$$\Longrightarrow \frac{\partial f}{\partial x} = 0$$

$$3x^2 - 3 = 0$$

$$3(x^2 - 1) = 0$$

$$x^2 - 1 = 0$$

$$\Longrightarrow x = \pm 1$$

$$\Longrightarrow \frac{\partial f}{\partial y} = 0$$

$$3y^2 - 12 = 0$$

$$3(y^2 - 4) = 0$$

$$y^2 - 4 = 0$$

$$\Longrightarrow y = \pm 2$$

$\therefore$ ***Stationary points*** : $(1,2), (-1,-2), (1,-2), (-1,2)$

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 6x$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6y$$

- **Step 3:**

(x,y)	r	s	t	$rt - s^2$	Conclusion	f at (a,b)
$(1,2)$	$6 > 0$	0	12	$72 > 0$	Minimum	2
$(-1,-2)$	$-6 < 0$	0	-12	$72 > 0$	Maximum	38
$(1,-2)$	6	0	-12	$-72 < 0$	Saddle point	---
$(-1,2)$	-6	0	12	$-72 < 0$	Saddle point	---

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Example 5: Examine $f(x,y) = x^3 + y^3 - 3axy$ for maximum & minimum value $(a > 0)$

Solution: $f(x,y) = x^3 + y^3 - 3axy$

- **Step 1:**

$$\Rightarrow \frac{\partial f}{\partial x} = 0$$

$$3x^2 - 3ay = 0$$

$$x^2 - ay = 0$$

$$y = \frac{x^2}{a} \ldots\ldots\ldots (1)$$

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

$$3y^2 - 3ax = 0$$

$$y^2 - ax = 0 \ldots\ldots\ldots (2)$$

$$Putting\ value\ of\ Eq(1)\ in\ Eq(2),$$

$$\left(\frac{x^2}{a}\right)^2 - ax = 0$$

$$x^4 - a^3x = 0$$

$$x(x^3 - a^3) = 0$$

$$\Rightarrow x = 0\,, x = a\,; put\ in\ Eq(1)$$

$$\Rightarrow\ y = 0\,, y = a$$

$\therefore$ ***Stationary points*** : $(0,0), (a,a)$

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 6x$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -3a$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6y$$

- **Step 3:**

(x,y)	r	s	t	$rt - s^2$	Conclusion	$f\ at\ (a,b)$
$(0,0)$	0	$-3a$	0	$-9a^2 < 0$	Saddle point	---
(a,a)	$6a > 0$	$-3a$	$6a$	$27a^2 > 0$	Minimum	$-a^3$

$\Rightarrow f_{min} = f_{(a,a)} = (a)^3 + (a)^3 - 3(a)^3 = -a^3$

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Example 6: Examine $f(x,y) = x^3 + y^3 - 3xy$ for maximum & minimum value.

Solution: $f(x,y) = x^3 + y^3 - 3xy$

- **Step 1:**

$$\Rightarrow \frac{\partial f}{\partial x} = 0$$

$$3x^2 - 3y = 0$$

$$x^2 - y = 0$$

$$y = x^2 \ldots\ldots\ldots (1)$$

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

$$3y^2 - 3x = 0$$

$$y^2 - x = 0 \ldots\ldots\ldots (2)$$

$$Putting\ value\ of\ Eq(1)\ in\ Eq(2),$$

$$(x^2)^2 - x = 0$$

$$x^4 - x = 0$$

$$x(x^3 - 1) = 0$$

$$\Rightarrow x = 0\,, x = 1\,; put\ in\ Eq(1)$$

$$\Rightarrow\ y = 0\,, y = 1$$

$\therefore$ ***Stationary points*** : $(0,0), (1,1)$

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 6x$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -3$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6y$$

- **Step 3:**

(x,y)	r	s	t	$rt - s^2$	Conclusion	$f\ at\ (a,b)$
(0,0)	0	-3	0	$-9 < 0$	Saddle point	---
(1,1)	$6 > 0$	-3	6	$27 > 0$	Minimum	-1

$\Rightarrow f_{min} = f_{(1,1)} = (1)^3 + (1)^3 - 3(1) = -1$

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Example 7: Discuss the maxima and minima of function $3x^2 - y^2 + x^3$.

Solution: $f(x,y) = 3x^2 - y^2 + x^3$

- **Step 1:**

$$\Rightarrow \frac{\partial f}{\partial x} = 0$$

$$6x + 3x^2 = 0$$

$$3x(2+x) = 0$$

$$\therefore x = 0 \quad or \quad x = -2$$

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

$$-2y = 0$$

$$\therefore y = 0$$

$\therefore$ ***Stationary points*** : $(0,0), (-2,0)$

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 6 + 6x$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$t = \frac{\partial^2 f}{\partial y^2} = -2$$

- **Step 3:**

(x,y)	r	s	t	$rt - s^2$	Conclusion	$f\ at\ (a,b)$
$(0,0)$	$6 > 0$	0	-2	$-12 < 0$	Saddle point	---
$(-2,0)$	$-6 < 0$	0	-2	$12 > 0$	Maximum	4

$\Rightarrow f_{max} = f_{(-2,0)} = 3(-2)^2 - 0 + (-2)^3 = 4$

Example 8: Find local extreme values of the function

$\boldsymbol{f(x,y) = 4x^2 + 9y^2 + 8x - 36y + 24}$

Solution: $f(x,y) = 4x^2 + 9y^2 + 8x - 36y + 24$

- **Step 1:**

$$\Rightarrow \frac{\partial f}{\partial x} = 0$$

$$8x + 8 = 0$$

$$8(x+1) = 0$$

$$(x+1) = 0$$

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$\therefore x = -1$

$\Longrightarrow \frac{\partial f}{\partial y} = 0$

$18y - 36 = 0$

$18(y - 2) = 0$

$y - 2 = 0$

$\therefore y = 2$

$\therefore$ ***Stationary points*** : $(-1,2)$

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 8$

  $s = \frac{\partial^2 f}{\partial x \partial y} = 0$

  $t = \frac{\partial^2 f}{\partial y^2} = 18$

- **Step 3:**

(x, y)	r	s	t	$rt - s^2$	Conclusion	f at (a, b)
$(-1,2)$	$8 > 0$	0	18	$144 > 0$	Minimum	-16

$\Longrightarrow f_{min} = f_{(-1,2)} = 4(-1)^2 + 9(2)^2 + 8(-1) - 36(2) + 24 = 4 + 36 - 8 - 72 + 24 \quad = -16$

Example 9: Find all local maxima, local minima and saddle points of the function

$\boldsymbol{f(x, y) = x^3 + 3xy^2 - 15x + y^3 - 15y}$.

Solution: $f(x, y) = x^3 + 3xy^2 - 15x + y^3 - 15y$

- **Step 1:**

$\Longrightarrow \frac{\partial f}{\partial x} = 0$

$3x^2 + 3y^2 - 15 = 0$

$3x^2 + 3y^2 = 15 \ldots\ldots\ldots (1)$

$\Longrightarrow \frac{\partial f}{\partial y} = 0$

$6xy + 3y^2 - 15 = 0$

$6xy + 3y^2 = 15 \ldots\ldots\ldots (2)$

Compare (1) & (2),

$\Longrightarrow 3x^2 + 3y^2 = 6xy + 3y^2$

$\Longrightarrow 3x^2 = 6xy$

$\Longrightarrow 3x^2 - 6xy = 0$

$\Longrightarrow 3x(x - 2y) = 0$

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$\Rightarrow x = 0 \; or \; x = 2y \ldots\ldots put \; in \; (1)$

When $x = 0$ $\Rightarrow From\ Eq(1) \Rightarrow 3y^2 = 15$

$$\Rightarrow 3(y^2 - 5) = 0$$

$$\Rightarrow y = \pm\sqrt{5}$$

$$\Rightarrow y = \sqrt{5}, -\sqrt{5}$$

When $x = 2y$ $\Rightarrow From\ Eq(1) \Rightarrow 12y^2 + 3y^2 = 15$

$$\Rightarrow 15y^2 = 1$$

$$\Rightarrow y^2 = 1$$

$$\Rightarrow y = \pm 1$$

$$\Rightarrow y = 1, -1$$

$\therefore$ ***Stationary points*** : $(0, \sqrt{5}), (0, -\sqrt{5}), (2,1), (-2, -1)$

- **Step 2:**

$$r = \frac{\partial^2 f}{\partial x^2} = 6x \; ,$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 6y \; ,$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6x + 6y$$

- **Step 3:**

(x, y)	r	s	t	$rt - s^2$	Conclusion
$(0, \sqrt{5})$	0	$6\sqrt{5}$	$6\sqrt{5}$	-180	Saddle point
$(0, -\sqrt{5})$	0	$-6\sqrt{5}$	$-6\sqrt{5}$	-180	Saddle point
$(2,1)$	$12 > 0$	6	18	$180 > 0$	Minimum
$(-2, -1)$	$-12 < 0$	-6	-18	$180 > 0$	Maximum

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H.W.

1) Discuss the maxima & minima of $f(x, y) = 2x^4 + y^2 - x^2 - 2y$

Ans: $f_{min} = -\frac{9}{8}$

2) Find all local maxima, local minima & saddle point of

$f(x, y) = x^2 + y^2 + 4x + 6y + 13$

Ans: $f_{min} = f_{(-2,-3)} = 0$

3) Show that the minimum value of $f(x,y) = xy + \frac{a^3}{x} + \frac{a^3}{y}$ is $3a^2$.

4) Find the extreme values of the function $x^3 + 3xy^2 - 3x^2 - 3y^2 + 7$

Ans: $f_{min} = f_{(2,0)} = 3$, $f_{max} = f_{(0,0)} = 7$, $(1,1)$ & $(1,-1)$ *are saddle points.*

5) Find extreme values of $f(x,y) = x^3 + y^3 - 63(x+y) + 12xy$

Ans: $f_{min} = f_{(3,3)} = -216$, $f_{max} = f_{(-7,-7)} = 784$,

$(5,-1)$ & $(-1,5)$ *are saddle points.*

6) Find local extreme values of $f(x,y) = xy - x^2 - y^2 - x$

Ans: f *has local maximum value at point* $\left(-\frac{2}{3}, -\frac{1}{3}\right)$

7) Examine the extreme values of $sinx + siny + \sin(x+y)$; $\left(0 \le x, y \le \frac{\pi}{2}\right)$

Ans: $f_{max} = f_{\left(\frac{\pi}{3}, \frac{\pi}{3}\right)} = \frac{3\sqrt{3}}{2}$

❖ Maximum & Minimum values with constrained variables:

- Sometimes we have to find the extreme values of a function of three (or more) variables $f(x,y,z)$ which are not independent but connected by some given relation $\emptyset(x,y,z) = 0$, The extreme values of $f(x,y,z)$ in such a situation are called constrained extreme values.
- In such situation, we use $\emptyset(x,y,z) = 0$ to eliminate one of the variables say z from the given function, thus converting the function as a function of only two variables & then find the extreme values of a function.

Example 1: Find the minimum value of $x^2 + y^2 + z^2$ with the constrained $x + y + z = 3a$.

Solution:

- $f(x,y,z) = x^2 + y^2 + z^2$ _____ (1)

  $\emptyset(x,y,z) = x + y + z - 3a = 0 \Longrightarrow z = 3a - x - y$ _____ (2)

- Substituting the values of z in equation (1),

  $f = x^2 + y^2 + (3a - x - y)^2$

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- Now applying maxima minima method
- **Step 1:**

$$\Rightarrow \frac{\partial f}{\partial x} = 0$$

$$2x - 2(3a - x - y) = 0$$

$$4x - 6a + 2y = 0$$

$$2x + y = 3a \quad _____ (3)$$

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

$$2y - 2(3a - x - y) = 0$$

$$2y - 6a + 2x + 2y$$

$$x + 2y = 3a \quad _____ (4)$$

Solving equation (3) & (4), $\Rightarrow x = y = a$

$\therefore$ ***Stationary point*** : (a, a)

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 2 + 2 = 4 > 0$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 2$$

$$t = \frac{\partial^2 f}{\partial y^2} = 4$$

- **Step 3:**

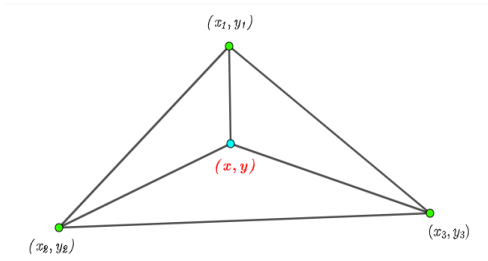
At point $(a, a) \Rightarrow rt - s^2 = 16 - 4 = 12 > 0$ & $r = 4 > 0$

$\therefore f$ has minimum value at (a, a).

$$\Rightarrow f_{min} = f_{(a,a)} = (a)^2 + (a)^2 + (3a - a - a)^2 = 3a^2$$

Example 2: Find a point within a triangle such that the sum of the squares of it's distance from the three vertices is a minimum.

Solution:



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- Let $(x_1, y_1), (x_2, y_2)$ & (x_3, y_3) be three vertices of a triangle. Let (x, y) be the interior point of the triangle.
- The required sum $= \left[\sqrt{(x-x_1)^2+(y-y_1)^2}\right]^2 + \left[\sqrt{(x-x_2)^2+(y-y_2)^2}\right]^2 + \left[\sqrt{(x-x_3)^2+(y-y_3)^2}\right]^2$
- $f(x,y,z) = (x-x_1)^2+(y-y_1)^2+(x-x_2)^2+(y-y_2)^2+(x-x_3)^2+(y-y_3)^2$
- Now applying maxima minima method
- **Step 1:**

$$\Rightarrow \frac{\partial f}{\partial x} = 0$$

$$2(x-x_1)+2(x-x_2)+2(x-x_3) = 0$$

$$(x-x_1)+(x-x_2)+(x-x_3) = 0$$

$$3x-(x_1+x_2+x_3) = 0$$

$$\therefore x = \frac{x_1+x_2+x_3}{3}$$

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

$$2(y-y_1)+2(y-y_2)+2(y-y_3) = 0$$

$$(y-y_1)+(y-y_2)+(y-y_3) = 0$$

$$3y-(y_1+y_2+y_3) = 0$$

$$\therefore y = \frac{y_1+y_2+y_3}{3}$$

$\therefore$ ***Stationary point*** : $(x,y) = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

- **Step 2:** $r = \frac{\partial^2 f}{\partial x^2} = 2+2+2 = 6 > 0$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$t = \frac{\partial^2 f}{\partial y^2} = 2+2+2 = 6$$

- **Step 3:**

At point (x, y)

$\Rightarrow rt - s^2 = 36 - 0 = 36 > 0$ & $r = 6 > 0$

$\therefore f$ has minimum value at $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$.

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❖ Lagrange Method of Undetermined Multipliers:

- This method is useful for finding the stationary points of the function subject to the given constraint with introduction of unknown constant λ known as Lagrange's undetermined multiplier.
- To find stationary value, consider a function three variables x, y, z with one condition

$$i.e\ u = f(x,y,z)\ with\ \phi(x,y,z) = 0$$

- **Working Rule:**

Step 1: Construct Lagrange's Function (Auxiliary equation),

$$F(x,y,z) = f(x,y,z) + \lambda\phi(x,y,z) = 0$$

Step 2: From the equation, write $\frac{dF}{dx} = 0, \frac{dF}{dy} = 0, \frac{dF}{dz} = 0$

Step 3: Solve these three equations along with $\phi(x,y,z) = 0$ &

Find the values of x, y, z & λ

Step 4: Put the value of $x, y, z\ in\ f(x,y,z)$.

Example 1: Find the minimum value of x^2yz^3 subject to constrain $2x + y + 3z = a$ using Lagrange's multiplier method.

Solution:

- Here $f(x,y,z) = x^2yz^3$ & $\emptyset(x,y,z) = 2x + y + 3z - a = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x,y,z) = f(x,y,z) + \lambda\phi(x,y,z) = 0$$
$$= x^2yz^3 + \lambda(2x + y + 3z - a) = 0$$

- **Step 2:**

$$\frac{\partial F}{\partial x} = 0 \Longrightarrow 2xyz^3 + 2\lambda = 0 \Longrightarrow \lambda = -xyz^3 \qquad \ldots\ldots(1)$$

$$\frac{\partial F}{\partial y} = 0 \Longrightarrow x^2z^3 + \lambda = 0 \Longrightarrow \lambda = -x^2z^3 \qquad \ldots\ldots(2)$$

$$\frac{\partial F}{\partial z} = 0 \Longrightarrow 3x^2yz^2 + 3\lambda = 0 \Longrightarrow \lambda = -x^2yz^2 \qquad \ldots\ldots(3)$$

- **Step 3:** From $(1), (2), (3)$,

$$xyz^3 = x^2z^3 = x^2yz^2$$

$$\Rightarrow yz = xz = xy$$

$$\therefore y = x\ and\ z = y$$

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- **Step 4:** Put $x = y = z$ in $2x + y + 3z = a$

$$\Rightarrow 2x + x + 3x = a$$

$$\Rightarrow 6x = a$$

$$\Rightarrow x = \frac{a}{6}, y = \frac{a}{6}, z = \frac{a}{6}$$

∴ Minimum value of the given function is:

$$x^2yz^3 = \left(\frac{a}{6}\right)^2\left(\frac{a}{6}\right)\left(\frac{a}{6}\right)^3 = \left(\frac{a}{6}\right)^6$$

Example 2: Determine the minimum value of x^2yz^2 subject to the condition $x + y + 2z = 5$ using method of Lagrange multipliers.

Solution:

- Here $f(x, y, z) = x^2yz^2$ & $\emptyset(x, y, z) = x + y + 2z - 5 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z) = 0$$
$$= x^2yz^2 + \lambda(x + y + 2z - 5) = 0$$

- **Step 2:**

$$\frac{\partial F}{\partial x} = 0 \Longrightarrow 2xyz^2 + \lambda = 0 \Longrightarrow \lambda = -2xyz^2 \qquad \ldots\ldots(1)$$

$$\frac{\partial F}{\partial y} = 0 \Longrightarrow x^2z^2 + \lambda = 0 \Longrightarrow \lambda = -x^2z^2 \qquad \ldots\ldots(2)$$

$$\frac{\partial F}{\partial z} = 0 \Longrightarrow 2x^2yz + 2\lambda = 0 \Longrightarrow \lambda = -x^2yz \qquad \ldots\ldots(3)$$

- **Step 3:** From (1), (2)

$$\Rightarrow -2xyz^2 = -x^2z^2$$

$$\Rightarrow 2y = x$$

$$\therefore y = \frac{x}{2}$$

- From (1), (3)

$$\Rightarrow -2xyz^2 = -x^2yz$$

$$\Rightarrow 2z = x$$

$$\therefore z = \frac{x}{2}$$

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- **Step 4:** Put $y = z = \frac{x}{2}$ in $x + y + 2z = 5$

$$\Rightarrow x + \frac{x}{2} + 2\left(\frac{x}{2}\right) = 5$$

$$\Rightarrow 2x + x + 2x = 10$$

$$\Rightarrow 5x = 10$$

$$\Rightarrow x = 2 \ \& \Rightarrow y = 1, z = 1$$

∴ Minimum value of the given function is:

$$x^2yz^2 = (2)^2(1)(1)^2 = 4$$

Example 3: The temperature T at a(x, y, z)in the space is $T = 400xyz^2$.Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$,using method of Lagrange's multipliers

Solution:

- Here $f(x, y, z) = 400xyz^2$ & $\emptyset(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z) = 0$$
$$= 400xyz^2 + \lambda(x^2 + y^2 + z^2 - 1) = 0$$

- **Step 2:**

$$\frac{\partial F}{\partial x} = 0 \Longrightarrow 400yz^2 + 2\lambda x = 0 \Longrightarrow \lambda = -200\frac{yz^2}{x} \qquad \ldots\ldots(1)$$

$$\frac{\partial F}{\partial y} = 0 \Longrightarrow 400xz^2 + 2\lambda y = 0 \Longrightarrow \lambda = -200\frac{xz^2}{y} \qquad \ldots\ldots(2)$$

$$\frac{\partial F}{\partial z} = 0 \Longrightarrow 800xyz + 2\lambda z = 0 \Longrightarrow \lambda = -400xy \qquad \ldots\ldots(3)$$

- **Step 3:** From (1)&(2) $\Rightarrow -200\frac{yz^2}{x} = -200\frac{xz^2}{y}$

$$\Rightarrow \frac{y}{x} = \frac{x}{y}$$

$$\Rightarrow x^2 = y^2$$

From (1)&(3) $\Rightarrow -200\frac{yz^2}{x} = -400xy$

$$\Rightarrow \frac{z^2}{x} = 2x$$

$$\Rightarrow z^2 = 2x^2$$

- **Step 4:** Put $x^2 = y^2$& $z^2 = 2x^2$ in $x^2 + y^2 + z^2 = 1$

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$\Rightarrow x^2 + x^2 + 2x^2 = 1$

$\Rightarrow 4x^2 = 1$

$\Rightarrow x^2 = \frac{1}{4}$

$\Rightarrow x = \pm\frac{1}{2}, y = \pm\frac{1}{2}$

But $\Rightarrow z^2 = 2x^2 \Rightarrow z^2 = \frac{1}{2} \Rightarrow z = \pm\frac{1}{\sqrt{2}}$

The points are $\left(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{\sqrt{2}}\right)$

∴ Highest temeperature on the surface of the sphere:

$T = 400xyz^2$

$= 400\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right)^2$

$= 50°C$

Example 4: Find numbers x, y & z such that $xyz = 8$ & $xy + yz + zx$ is minimum using Lagrange's method of undetermined multiplier.

Solution:

- Here $f(x,y,z) = xy + yz + zx$ & $\emptyset(x,y,z) = xyz - 8 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x,y,z) = f(x,y,z) + \lambda\phi(x,y,z) = 0$$
$$= xy + yz + zx + \lambda(xyz - 8) = 0$$

- **Step 2:**

$\frac{\partial F}{\partial x} = 0 \Longrightarrow y + z + \lambda yz = 0 \Longrightarrow \lambda = -\frac{(y+z)}{yz}$ .....(1)

$\frac{\partial F}{\partial y} = 0 \Longrightarrow x + z + \lambda xz = 0 \Longrightarrow \lambda = -\frac{(x+z)}{xz}$ .....(2)

$\frac{\partial F}{\partial z} = 0 \Longrightarrow y + x + \lambda xy = 0 \Longrightarrow \lambda = -\frac{(y+x)}{xy}$ .....(3)

- **Step 3:** From (1) &(2), $-\frac{(y+z)}{yz} = -\frac{(x+z)}{xz}$

$$\Rightarrow x(y+z) = y(x+z)$$
$$\Rightarrow xy + xz = xy + yz$$
$$\Rightarrow xz = yz$$
$$\Rightarrow x = y$$

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From (2) &(3), $-\frac{(y+z)}{yz} = --\frac{(y+x)}{xy}$

$$\Rightarrow x(y+z) = z(y+x)$$

$$\Rightarrow xy + xz = yz + xz$$

$$\Rightarrow xy = yz$$

$$\Rightarrow x = z$$

$\therefore x = y = z$

- **Step 4:** Put $x = y = z$ in $xyz = 8$

$$\Rightarrow x.x.x = 8 \Rightarrow x^3 = 8$$

$$\Rightarrow x = 2, y = 2, z = 2$$

Example 5: Find the shortest and longest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$.

Solution: Let $p(x, y, z)$ be any point on the sphere $x^2 + y^2 + z^2 = 24$ & distance from $(1,2,-1)$ to (x, y, z) is,

$$d = \sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}$$

- Here $f(x, y, z) = (x-1)^2 + (y-2)^2 + (z+1)^2$ & $\emptyset(x, y, z) = x^2 + y^2 + z^2 - 24 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z) = 0$$

$$= (x-1)^2 + (y-2)^2 + (z+1)^2 + \lambda(x^2 + y^2 + z^2 - 24) = 0$$

- **Step 2:**

$$\frac{\partial F}{\partial x} = 0 \Longrightarrow 2(x-1) + 2\lambda x = 0 \Longrightarrow (x-1) + \lambda x = 0 \Longrightarrow \lambda = -\frac{(x-1)}{x} = -1 + \frac{1}{x} \ldots(1)$$

$$\frac{\partial F}{\partial y} = 0 \Longrightarrow 2(y-2) + 2\lambda y = 0 \Longrightarrow (y-2) + \lambda y = 0 \Longrightarrow \lambda = -\frac{(y-2)}{y} = -1 + \frac{2}{y} \ldots (2)$$

$$\frac{\partial F}{\partial z} = 0 \Longrightarrow 2(z+1) + 2\lambda z = 0 \Longrightarrow (z+1) + \lambda z = 0 \Longrightarrow \lambda = -\frac{(z+1)}{z} = -1 - \frac{1}{z} \ldots (3)$$

- **Step 3:** From (1)&(2), $-1 + \frac{1}{x} = -1 + \frac{2}{y}$

$$\Rightarrow \frac{1}{x} = \frac{2}{y}$$

$$\Rightarrow y = 2x$$

From (1) &(3), $-1 + \frac{1}{x} = -1 - \frac{1}{z}$

$$\Rightarrow \frac{1}{x} = -\frac{1}{z}$$

$$\Rightarrow z = -x$$

- **Step 4:** Put value of x & y in $x^2 + y^2 + z^2 = 24$

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$\Rightarrow (x)^2 + (2x)^2 + (-x)^2 = 24$

$\Rightarrow x^2 + 4x^2 + x^2 = 24$

$\Rightarrow 6x^2 = 24$

$\Rightarrow x^2 = 4$

$\Rightarrow x = \pm 2 \ \& \ y = \pm 4, z = \mp 2$

For$(x, y, z) = (-2, -4, 2) \Rightarrow Farthest\ Point$

$\Rightarrow D^2 = 54$

$\Rightarrow D = \sqrt{54}$ =7.35 **(Longest Distance)**

- For$(x, y, z) = (2, 4, -2) \Rightarrow Nearest\ Point$

  $\Rightarrow D^2 = 6$

  $\Rightarrow \mathrm{D} = \sqrt{6} = 2.45$ **(Shortest Distance)**

Example 6 : Find the shortest distance from the point $(1, 2, 2)$ on the surface of the sphere $x^2 + y^2 + z^2 = 16$.

Solution: Let $p(x, y, z)$ be any point on the sphere $x^2 + y^2 + z^2 = 16$ & distance from (1,2,2) to (x, y, z) is,

$$d = \sqrt{(x-1)^2 + (y-2)^2 + (z-2)^2}$$

- Here $f(x, y, z) = (x-1)^2 + (y-2)^2 + (z-2)^2$ & $\emptyset(x, y, z) = x^2 + y^2 + z^2 - 16 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z) = 0$$
$$= (x-1)^2 + (y-2)^2 + (z-2)^2 + \lambda(x^2 + y^2 + z^2 - 16) = 0$$

- **Step 2:**

$\frac{\partial F}{\partial x} = 0 \Longrightarrow 2(x-1) + 2\lambda x = 0 \Longrightarrow (x-1) + \lambda x = 0 \Longrightarrow \lambda = -\frac{(x-1)}{x} = -1 + \frac{1}{x}$ ...(1)

$\frac{\partial F}{\partial y} = 0 \Longrightarrow 2(y-2) + 2\lambda y = 0 \Longrightarrow (y-2) + \lambda y = 0 \Longrightarrow \lambda = -\frac{(y-2)}{y} = -1 + \frac{2}{y}$ ... (2)

$\frac{\partial F}{\partial z} = 0 \Longrightarrow 2(z-2) + 2\lambda z = 0 \Longrightarrow (z-1) + \lambda z = 0 \Longrightarrow \lambda = -\left(\frac{z-2}{z}\right) = -1 + \frac{2}{z}$ ... (3)

- **Step 3:** From (1)&(2), $-1 + \frac{1}{x} = -1 + \frac{2}{y}$

$$\Rightarrow \frac{1}{x} = \frac{2}{y}$$

$$\Rightarrow y = 2x$$

From (1) &(3), $-1 + \frac{1}{x} = -1 + \frac{2}{z}$

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$$\Rightarrow \frac{1}{x} = \frac{2}{z}$$

$$\Rightarrow z = 2x$$

- **Step 4:** Put value of x & y in $x^2 + y^2 + z^2 = 16$

$$\Rightarrow (x)^2 + (2x)^2 + (2x)^2 = 16$$

$$\Rightarrow x^2 + 4x^2 + 4x^2 = 16$$

$$\Rightarrow 9x^2 = 16$$

$$\Rightarrow x^2 = \frac{16}{9}$$

$$\Rightarrow x = \pm\frac{4}{3} \ \& \ y = \pm\frac{8}{3}, z = \pm\frac{8}{3}$$

- For$(x, y, z) = \left(\frac{4}{3}, \frac{8}{3}, \frac{8}{3}\right) \Rightarrow Nearest\ Point$

$$\Rightarrow D^2 = \left(\frac{4}{3} - 1\right)^2 + \left(\frac{8}{3} - 2\right)^2 + \left(\frac{8}{3} - 2\right)^2$$

$$= \frac{1}{9} + \frac{4}{9} + \frac{4}{9}$$

$$= 1$$

$\Rightarrow \boldsymbol{D} = \sqrt{\mathbf{1}}$ **=1** **(Shortest Distance)**

- For$(x, y, z) = \left(-\frac{4}{3}, -\frac{8}{3}, -\frac{8}{3}\right) \Rightarrow Farthest\ Point$

$$\Rightarrow D^2 = \left(-\frac{4}{3} - 1\right)^2 + \left(-\frac{8}{3} - 2\right)^2 + \left(-\frac{8}{3} - 2\right)^2$$

$$= \frac{49}{9} + \frac{196}{9} + \frac{196}{9} = \frac{441}{9} = 49$$

$\Rightarrow D = \sqrt{49}$ =7

Example 7: Find the maximum and minimum values of the function

$\boldsymbol{f(x, y) = 3x + 4y}$ **on the circle** $x^2 + y^2 = 1$

- Here $f(x, y, z) = 3x + 4y$ & $\emptyset(x, y, z) = x^2 + y^2 - 1 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z) = 0$$

$$= (3x + 4y) + \lambda(x^2 + y^2 - 1) = 0$$

- **Step 2:**

$$\frac{\partial F}{\partial x} = 0 \Longrightarrow 3 + \lambda(2x) = 0 \Longrightarrow x = \frac{3}{2\lambda} \qquad \ldots\ldots(1)$$

$$\frac{\partial F}{\partial y} = 0 \Longrightarrow 4 + \lambda(2\lambda) = 0 \Longrightarrow y = \frac{2}{\lambda} \qquad \ldots\ldots(2)$$

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- **Step 3:** put value from (1), (2) in $x^2 + y^2 - 1 = 0$

$$\Rightarrow \left(\frac{3}{2\lambda}\right)^2 + \left(\frac{2}{\lambda}\right)^2 = 1$$

$$\Rightarrow \frac{9}{4\lambda^2} + \frac{4}{\lambda^2} = 1$$

$$\Rightarrow \frac{9+16}{4\lambda^2} = 1$$

$$\Rightarrow \frac{25}{4\lambda^2} = 1$$

$$\therefore \lambda = \pm\frac{5}{2}$$

- For $\lambda = \frac{5}{2} \Rightarrow x = \frac{3}{2\left(\frac{5}{2}\right)} = \frac{3}{5}$ & $x = \frac{2}{\left(\frac{5}{2}\right)} = \frac{4}{5}$

  $\Rightarrow$At Point: $(x, y) = \left(\frac{3}{5}, \frac{4}{5}\right)$

  $\Rightarrow f(x, y) = f\left(\frac{3}{5}, \frac{4}{5}\right) = 3\left(\frac{3}{5}\right) + 4\left(\frac{4}{5}\right) = \mathbf{5}$ (Maximum value)

- For $\lambda = -\frac{5}{2} \Rightarrow x = \frac{3}{2\left(-\frac{5}{2}\right)} = -\frac{3}{5}$ & $x = \frac{2}{\left(-\frac{5}{2}\right)} = -\frac{4}{5}$

  $\Rightarrow$At Point: $(x, y) = \left(-\frac{3}{5}, -\frac{4}{5}\right)$

  $\Rightarrow f(x, y) = f\left(-\frac{3}{5}, -\frac{4}{5}\right) = 3\left(-\frac{3}{5}\right) + 4\left(-\frac{4}{5}\right) = \mathbf{-5}$ (Minimum value)

Example 8: Find the point upon the plane $2x + 3y - z = 5$ which is nearest to the origin, using Lagrange's method of undetermined multiplier.

Solution: Let $p(x, y, z)$ be any point on the plane $2x + 3y - z = 5$ & distance from (0,0,0) to (x, y, z) is,

$$d = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2}$$

- Here $f(x, y, z) = d^2 = x^2 + y^2 + z^2$ & $\emptyset(x, y, z) = 2x + 3y - z - 5 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z) = 0$$
$$= x^2 + y^2 + z^2 + \lambda(2x + 3y - z - 5) = 0$$

- **Step 2:**

$$\frac{\partial F}{\partial x} = 0 \Longrightarrow 2x + 2\lambda = 0 \Longrightarrow x = -\lambda \quad \ldots\ldots(1)$$

$$\frac{\partial F}{\partial y} = 0 \Longrightarrow 2y + 3\lambda = 0 \Longrightarrow y = -\frac{3}{2}\lambda \quad \ldots\ldots(2)$$

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$\frac{\partial F}{\partial z} = 0 \Longrightarrow 2z - \lambda = 0 \Longrightarrow z = \frac{\lambda}{2}$ ... ... (3)

- **Step 3:** Put the value of x, y, z in equation of plane.

$\Rightarrow 2x + 3y - z = 5$

$\Rightarrow 2(-\lambda) + 3\left(-\frac{3}{2}\lambda\right) - \left(\frac{\lambda}{2}\right) = 5$

$\Rightarrow -4\lambda - 9\lambda - \lambda = 10$

$\Rightarrow -14\lambda = 10$

$\Rightarrow \lambda = -\frac{10}{14}$

$\Rightarrow \lambda = -\frac{5}{7}$, $Put\ in\ (1), (2), (3)$

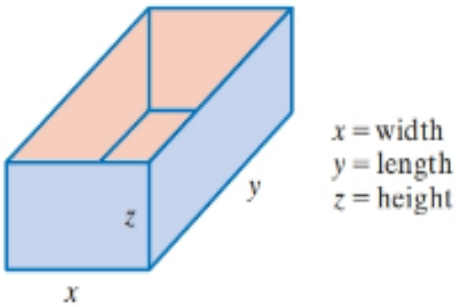
Step 4: Put value of $\lambda\ \ in\ (1), (2), (3)$

$\Rightarrow x = \frac{5}{7}, y = \frac{15}{14}, z = -\frac{5}{14}$

$\Rightarrow$ Hence the required nearest point from origin is $(x, y, z) = \left(\frac{5}{7}, \frac{15}{14}, -\frac{5}{14}\right)$.

Example 9: A rectangular box open at the top is to have a volume of 32 cubic units. Find the dimensions of the box requiring least material for its construction.

Solution:



- Let x, y & z be the dimension of the box.
- The box is open at the top,

$\therefore Surface\ area = S = xy + 2yz + 2zx$ ... ... (1)

$\therefore Volume = V = xyz = 32$ ... ... (2)

- Let $f(x, y, z) = xy + 2yz + 2zx$ & $\emptyset(x, y, z) = xyz - 32 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z) = 0$$
$$= xy + 2yz + 2zx + \lambda(xyz - 32) = 0$$

- **Step 2:**

$$\frac{\partial F}{\partial x} = 0 \Longrightarrow y + 2z + \lambda yz = 0 \Longrightarrow \lambda = -\frac{(y+2z)}{yz} = -\frac{1}{y} - \frac{2}{y} \qquad \ldots\ldots(1)$$

$$\frac{\partial F}{\partial y} = 0 \Longrightarrow x + 2z + \lambda xz = 0 \Longrightarrow \lambda = -\frac{(x+2z)}{xz} = -\frac{1}{z} - \frac{2}{x} \qquad \ldots\ldots(2)$$

$$\frac{\partial F}{\partial z} = 0 \Longrightarrow 2y + 2x + \lambda xy = 0 \Longrightarrow \lambda = -\frac{(2y+2x)}{xy} = -\frac{2}{y} - \frac{2}{x} \qquad \ldots\ldots(3)$$

- **Step 3:** From (1) &(2), $-\frac{1}{y} - \frac{2}{y} = -\frac{1}{z} - \frac{2}{x}$

$$\Rightarrow \frac{2}{y} = \frac{2}{x}$$

$$\Rightarrow x = y$$

From (1) &(3), $-\frac{1}{y} - \frac{2}{y} = -\frac{2}{y} - \frac{2}{x}$

$$\Rightarrow \frac{1}{z} = \frac{2}{x}$$

$$\Rightarrow z = \frac{x}{2}$$

- **Step 4:** Put $x = y = z$ in $xyz = 32$

$$\Rightarrow x.x.\frac{x}{2} = 32$$

$$\Rightarrow x^3 = 64$$

$$\Rightarrow x = 4\,, y = 4\,, z = \frac{4}{2} = 2$$

Hence, dimensions of box requiring least material for its construction are 4,4,2.

Example 10: A rectangular box without a lid is to be made from $12m^2$ cardboard. Find the maximum volume of such a box.

Solution:

- Let x, y & z be the dimension of the box.
- The box is without lid,
- $\therefore Volume = V = xyz \quad \ldots\ldots\ldots(1)$

  $\therefore Surface\ area = S = xy + 2yz + 2zx = 12 \quad \ldots\ldots\ldots(2)$
- Let $f(x,y,z) = xyz$ & $\emptyset(x,y,z) = xy + 2yz + 2zx - 12 = 0$
- **Step 1:** Construct Lagrange's Function

$$F(x,y,z) = f(x,y,z) + \lambda\phi(x,y,z) = 0$$

$$= xy + 2yz + 2zx + \lambda(xyz - 32) = 0$$

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- **Step 2:**

$$\frac{\partial F}{\partial x} = 0 \Longrightarrow yz + \lambda(y+2z) = 0 \Longrightarrow \lambda = -\frac{yz}{(y+2z)} \Longrightarrow \frac{1}{\lambda} = -\frac{(y+2z)}{yz} = -\frac{1}{z} - \frac{2}{y} \quad \ldots\ldots(1)$$

$$\frac{\partial F}{\partial y} = 0 \Longrightarrow xz + \lambda(x+2z) = 0 \Longrightarrow \lambda = -\frac{xz}{(x+2z)} \Longrightarrow \frac{1}{\lambda} = -\frac{(x+2z)}{xz} = -\frac{1}{z} - \frac{2}{x} \quad \ldots\ldots(2)$$

$$\frac{\partial F}{\partial z} = 0 \Longrightarrow xy + \lambda(2y+2x) = 0 \Longrightarrow \lambda = -\frac{xy}{(2y+2x)} \Longrightarrow \frac{1}{\lambda} = -\frac{(2y+2x)}{xy} = -\frac{2}{y} - \frac{2}{x} \quad \ldots\ldots(3)$$

- **Step 3:** From (1) &(2), $-\frac{1}{y} - \frac{2}{y} = -\frac{1}{z} - \frac{2}{x}$

$$\Rightarrow \frac{2}{y} = \frac{2}{x}$$

$$\Rightarrow x = y$$

From (1) &(3), $-\frac{1}{y} - \frac{2}{y} = -\frac{2}{y} - \frac{2}{x}$

$$\Rightarrow \frac{1}{z} = \frac{2}{x}$$

$$\Rightarrow z = \frac{x}{2}$$

- **Step 4:** Put $x = y = z$ in $xy + 2yz + 2zx = 12$

$$\Rightarrow x.x + 2.x.\frac{x}{2} + 2\frac{x}{2}.x = 12$$

$$\Rightarrow 3x^2 = 12$$

$$\Rightarrow x^2 = 4$$

$$\Rightarrow x = 2, y = 2, z = 1$$

- Hence maximum volume $v = xyz = (2)(2)(1) = 4\ m^3$

H.W.

1) Find the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest and farthest from the point $(3,1,-1)$.

Ans: Closest point $= \left(\frac{6}{\sqrt{11}}, \frac{2}{\sqrt{11}}, -\frac{2}{\sqrt{11}}\right) \Rightarrow d = 1.32$

Farthest point $= \left(-\frac{6}{\sqrt{11}}, -\frac{2}{\sqrt{11}}, \frac{2}{\sqrt{11}}\right) \Rightarrow d = 5.32$

2) Find the point on the plane $x + 2y + 3z = 13$ closest to the point $(1,1,1)$

Ans: $\left(\frac{3}{2}, 2, \frac{5}{2}\right)$

3) Find the number $x, y and z$ such that $xyz = 16$ and $18xy + 16yz + 12xz$ is minimum, using the Lagrange's method of undetermined coefficients.

Ans:Minimum point : $\left(\frac{8}{3}, 2, 3\right)$

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4) Find the greatest & least values that the function $f(x,y) = xy$ takes on the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$

Ans: Greatest value $= 2$ & least value $= -2$

5) Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.

6) A rectangular box opens the top is to have a volume of 108 c.c. find the dimension of the box requiring least material for its construction.

Ans: $(\mathbf{6}, \mathbf{6}, \mathbf{3})$

❖ Gradient & Directional Derivatives:

VECTOR CALCULUS → **Vector Differentiation** → **Scalar point function** → **Gradient & Directional Derivatives**

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Gradient:

- If $\emptyset(x,y,z)$ be any Scalar point function then, the Gradient of $\emptyset$ is denoted by $grad\ \emptyset$ (or) $\nabla\emptyset$,which is defined as

$$\boldsymbol{grad\ \emptyset = \nabla\emptyset = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)\emptyset = \left(i\frac{\partial\emptyset}{\partial x} + j\frac{\partial\emptyset}{\partial y} + k\frac{\partial\emptyset}{\partial z}\right)}$$

- For Example,
- If $f = x^2 + y^2 + z^2$ then $\nabla f = 2xi + 2yj + 2zk$.
- If $f = xyz$ then $\nabla f = (yz)i + (xz)j + (xy)k$

Directional derivatives:

- Let $\emptyset(x,y,z)$ be scalar point function then $\frac{\partial\emptyset}{\partial x}, \frac{\partial\emptyset}{\partial y}, \frac{\partial\emptyset}{\partial z}$ are directional derivative of $\emptyset$ in the direction of the co-ordinate axis
- The directional derivatives of a scalar point function $\emptyset(x,y,z)$ at point p in direction of vector $\vec{a}$,

$$\therefore\ \boldsymbol{D.D.} = (\boldsymbol{\nabla\emptyset})_p . \hat{\boldsymbol{a}} = (\boldsymbol{\nabla\emptyset})_p \cdot \frac{\vec{a}}{|\vec{a}|} \quad (Here\ \hat{a} = a_1\hat{\imath} + a_2\hat{\imath} + a_3\hat{\imath}\ \ is\ Unit\ vector)$$

Example 1: Find $grad\ \emptyset$ if $\emptyset = 3x^2y - y^3\ z^2$ at $(1, -2, 1)$.

Solution: By definition of gradient vector,

- $grad\ \emptyset = \nabla\emptyset = i\frac{\partial\emptyset}{\partial x} + j\frac{\partial\emptyset}{\partial y} + k\frac{\partial\emptyset}{\partial z}$

$$= (6xy - 0)i + (3x^2 - 3y^2z^2)j + (0 - 2y^3z)k$$

$\therefore (grad\ \emptyset)p = (\nabla\emptyset)p = (\nabla\emptyset)(1, -2, 1) = -12i - 9j + 16k.$

Example 2: Find $grad\ \emptyset$ if $\emptyset = log(x^2 + y^2 + z^2)$ at $(1, 2, 1)$.

Solution: By definition of gradient vector

- $grad\ \emptyset = \nabla\emptyset = i\frac{\partial\emptyset}{\partial x} + j\frac{\partial\emptyset}{\partial y} + k\frac{\partial\emptyset}{\partial z}$

$= \frac{\partial log(x^2+y^2+z^2)}{\partial x} i + \frac{\partial log(x^2+y^2+z^2)}{\partial y} j + \frac{\partial log(x^2+y^2+z^2)}{\partial z} k$

$$= \frac{2x}{x^2 + y^2 + z^2} i + \frac{2y}{x^2 + y^2 + z^2} j + \frac{2z}{x^2 + y^2 + z^2} k$$

$\therefore grad\ \emptyset = \frac{2}{x^2+y^2+z^2}\ (xi + yj + zk)$

$\therefore (grad\ \emptyset)p = (\nabla\emptyset)p = \frac{1}{3}(1i + 2j + 1k) = \frac{1}{3}i + \frac{2}{3}j + \frac{1}{3}k$

Example 3: The temperature at any point in space is given by $T = xy + yz + zx$.

Determine the derivative of T in the direction of vector $3i - 4k$ at (1,1,1).

Solution:

- D.D. = $(\nabla T)_p \cdot \frac{\vec{a}}{|\vec{a}|}$

By definition of Gradient vector

- $grad\ T = \nabla T = \frac{\partial T}{\partial x} i + \frac{\partial T}{\partial y} j + \frac{\partial T}{\partial z} k = (y + z)i + (x + z)j + (y + x)k$

$at\ point\ p(x, y, z) = (1,1,1)$

$$\therefore (grad\ T)p = (\nabla T)_p = 2i + 2j + 2k$$

- Direction vector $\vec{a} = 3i - 4k$ then $|\vec{a}| = \sqrt{3^2 + 0^2 + (-4)^2} = \sqrt{25} = 5$
- D.D. = $(\nabla T)_p \cdot \frac{\vec{a}}{|\vec{a}|}$

$= (2i + 2j + 2k).\left(\frac{3i-4k}{5}\right) = \frac{1}{5}(6 - 8) = \frac{-2}{5}$

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Example 4: Find the directional derivative of $\emptyset = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ in the direction of vector $2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}$.

Solution:

By definition of Gradient vector

- $grad\ \emptyset = \nabla\emptyset = \frac{\partial\emptyset}{\partial x}i + \frac{\partial\emptyset}{\partial y}j + \frac{\partial\emptyset}{\partial z}k$

$$= (4z^3 - 6xy^2z)\hat{\imath} + (-6x^2yz)\hat{\jmath} + (12xz^2 - 3x^2y^2)\hat{k}$$

$$\therefore (grad\ \emptyset)p = (\nabla\emptyset)_{(2,-1,2)} = 8\hat{\imath} + 48\hat{\jmath} + 84\hat{k}$$

- Direction vector $\vec{a} = 2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}$ then $|\vec{a}| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = 7$
- D.D. = $(\nabla\emptyset)_p \cdot \frac{\vec{a}}{|\vec{a}|} = (8\hat{\imath} + 48\hat{\jmath} + 84\hat{k}).\left(\frac{2\hat{\imath}+3\hat{\jmath}+6\hat{k}}{7}\right) = \frac{16+144+504}{7} = \frac{664}{7}$

Example 5: Find the directional derivative of $x^2y^2z^2$ at $(1, 1, -1)$ along a direction equally inclined with co-ordinate axis.

Solution:

- In the direction equally inclined with co-ordinate axes is $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$

By definition of Gradient vector,

- $grad\ \emptyset = \nabla\emptyset = \frac{\partial\emptyset}{\partial x}i + \frac{\partial\emptyset}{\partial y}j + \frac{\partial\emptyset}{\partial z}k$

$$= (2x.y^2z^2)\hat{\imath} + (2y.y^2z^2)\hat{\jmath} + (2z.x^2y^2)\hat{k}$$

$$\therefore (grad\ \emptyset)p = (\nabla\emptyset)_{(1,1,-1)} = 2\hat{\imath} + 2\hat{\jmath} - 2\hat{k}$$

- Direction vector $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$ then $|\vec{a}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$
- D.D. = $(\nabla\emptyset)_p \cdot \frac{\vec{a}}{|\vec{a}|}$

$$= (2\hat{\imath} + 2\hat{\jmath} - 2\hat{k}).\left(\frac{\hat{\imath}+\hat{\jmath}+\hat{k}}{\sqrt{3}}\right) = \frac{2+2-2}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

Example 6: Find the directional derivative of $f(x, y, z) = x^3 - xy^2 - z$ at (1,1,0) in the direction of $2i - 3j + 6k$.

Solution: By definition of Gradient vector

- $grad\ f = \nabla f = \frac{\partial f}{\partial x}i + \frac{\partial f}{\partial y}j + \frac{\partial f}{\partial z}k$

$$= (3x^2 - y^2)\hat{\imath} + (-2xy)\hat{\jmath} + (-1)\hat{k}$$

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$\therefore (grad\ f)p = (\nabla f)_{(1,1,0)} = 2\hat{\imath} - 2\hat{\jmath} - \hat{k}$

- Direction vector $\vec{a} = 2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}$ then $|\vec{a}| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = 7$
- D.D. = $(\nabla f)_p \cdot \frac{\vec{a}}{|\vec{a}|} = (2\hat{\imath} - 2\hat{\jmath} - \hat{k}).\left(\frac{2\hat{\imath}-3\hat{\jmath}+6\hat{k}}{7}\right) = \frac{4+6-6}{7} = \frac{4}{7}$

Example 7: Find the directional derivative of $\emptyset(x, y, z) = xy^2 + yz^2$ at the point $(2, -1, 1)$ in the direction of vector $i + 2j + 2k$.

Solution: By definition of Gradient vector

- $grad\ \emptyset = \nabla\emptyset = \frac{\partial\emptyset}{\partial x}i + \frac{\partial\emptyset}{\partial y}j + \frac{\partial\emptyset}{\partial z}k$

$= (y^2)\hat{\imath} + (2xy + z^2)\hat{\jmath} + (2yz)\hat{k}$

$\therefore (grad\ \emptyset)p = (\nabla\emptyset)_{(2,-1,1)} = \hat{\imath} - 3\hat{\jmath} - 2\hat{k}$

- Direction vector $\vec{a} = \hat{\imath} + 2\hat{\jmath} + 2\hat{k}$ then $|\vec{a}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$
- D.D. = $(\nabla\emptyset)_p \cdot \frac{\vec{a}}{|\vec{a}|} = (\hat{\imath} - 3\hat{\jmath} - 2\hat{k}).\left(\frac{\hat{\imath}+2\hat{\jmath}+2\hat{k}}{3}\right) = \frac{1-6-4}{3} = \frac{-9}{3} = -3$

Example 8: Find the directional derivative of $f(x, y, z) = xyz$ at the point $P(-1, 1, 3)$ in the direction the vector $\overline{a} = \hat{\imath} - 2\hat{\jmath} + 2\hat{k}$

Solution: By definition of Gradient vector

- $grad\ f = \nabla f = \frac{\partial f}{\partial x}i + \frac{\partial f}{\partial y}j + \frac{\partial f}{\partial z}k$

$= (yz)\hat{\imath} + (xz)\hat{\jmath} + (xy)\hat{k}$

$\therefore (grad\ f)p = (\nabla f)_{(-1,1,3)} = 3\hat{\imath} - 3\hat{\jmath} - \hat{k}$

- Direction vector $\vec{a} = \hat{\imath} - 2\hat{\jmath} + 2\hat{k}$ then $|\vec{a}| = \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{9} = 3$
- D.D. = $(\nabla f)_p \cdot \frac{\vec{a}}{|\vec{a}|} = (3\hat{\imath} - 3\hat{\jmath} - \hat{k}).\left(\frac{\hat{\imath}-2\hat{\jmath}+2\hat{k}}{3}\right) = \frac{3+6-2}{3} = \frac{7}{3}$

Example 9: Find the directional derivative of $f(x, y) = x^2 sin2y$ at the point $(1, \frac{\pi}{2})$ in the direction of $v = 3\hat{\imath} - 4\hat{\jmath}$

Solution: By definition of Gradient vector

- $grad\ f = \nabla f = \frac{\partial f}{\partial x}i + \frac{\partial f}{\partial y}j$

$= (2x.sin2y)\hat{\imath} + (2x^2.cos2y)\hat{\jmath}$

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$\therefore (grad\, f)p = (\nabla f)_{\left(1,\frac{\pi}{2}\right)} = 0\hat{\imath} - 2\hat{\jmath} = -2\hat{\jmath}$

- Direction vector $\vec{a} = 3\hat{\imath} - 4\hat{\jmath}$ then $|\vec{a}| = \sqrt{9^2 + 16^2} = \sqrt{25} = 5$
- D.D. = $(\nabla f)_p \cdot \frac{\vec{a}}{|\vec{a}|} = (-2\hat{\jmath}).\left(\frac{3\hat{\imath}-4\hat{\jmath}}{5}\right) = \frac{8}{5}$

Example 10: If vector $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ then show that $\nabla r^n = nr^{n-2}.(Vector\ r)$

Solution:

- $r^2 = x^2 + y^2 + z^2$
- Firstly, differentiating r partially with respect to x, y & z respectively, we get

$$\Rightarrow 2r\frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r} \quad \text{Similarly,} \frac{\partial r}{\partial y} = \frac{y}{r} \ \& \ \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\nabla r^n = \hat{\imath}\frac{\partial}{\partial x}(r^n) + \hat{\jmath}\frac{\partial}{\partial y}(r^n) + \hat{k}\frac{\partial}{\partial z}(r^n)$$

$$= \hat{\imath}\left[nr^{n-1}\frac{\partial r}{\partial x}\right] + \hat{\jmath}\left[nr^{n-1}\frac{\partial r}{\partial y}\right] + \hat{k}\left[nr^{n-1}\frac{\partial r}{\partial z}\right]$$

$$= \hat{\imath}\left[nr^{n-1}\frac{x}{r}\right] + \hat{\jmath}\left[nr^{n-1}\frac{x}{r}\right] + \hat{k}\left[nr^{n-1}\frac{z}{r}\right]$$

$$= \frac{nr^{n-1}}{r}\left(x\hat{\imath} + y\hat{\jmath} + z\hat{k}\right)$$

$$= nr^{n-2}.(\vec{r})$$

$$= nr^{n-2}.(Vector\ r)$$

Example 11: Let $f = \ln r$, where $\bar{r} = x\,\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and $r = |\bar{r}|$. Find $Grad\ f$.

- $r^2 = x^2 + y^2 + z^2$
- Firstly, differentiating r partially with respect to x, y & z respectively, we get

$$\Rightarrow 2r\frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r} \quad \text{Similarly,} \frac{\partial r}{\partial y} = \frac{y}{r} \ \& \ \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\nabla lnr = \hat{\imath}\frac{\partial}{\partial x}(lnr) + \hat{\jmath}\frac{\partial}{\partial y}(lnr) + \hat{k}\frac{\partial}{\partial z}(lnr)$$

$$= \hat{\imath}\left[\frac{1}{r}\frac{\partial r}{\partial x}\right] + \hat{\jmath}\left[n\frac{1}{r}\frac{\partial r}{\partial y}\right] + \hat{k}\left[n\frac{1}{r}\frac{\partial r}{\partial z}\right]$$

$$= \hat{\imath}\left[\frac{1}{r}\frac{x}{r}\right] + \hat{\jmath}\left[\frac{1}{r}\frac{x}{r}\right] + \hat{k}\left[\frac{1}{r}\frac{z}{r}\right]$$

$$= \frac{1}{r^2}\left(x\hat{\imath} + y\hat{\jmath} + z\hat{k}\right)$$

$$= \frac{1}{r^2}.(\vec{r})$$

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Example 12: Find the gradient of $f(x,y,z) = 2z^3 - 3(x^2+y^2)z + tan^{-1}(xz)$ at (1,1,1**)**

Ans : $\left(-\frac{11}{2}\right)\hat{\imath} - 6\hat{\jmath} + \left(\frac{1}{2}\right)\hat{k}$

Example 13: Find the directional derivative of $\emptyset(x,y,z) = 3e^x cos(yz)$ at point the (0,0,0) in the direction of vector $2i + j - 2k$

Ans :2

Example 14: Find the derivative of $f(x,y) = x^2 + xy + y^2$ in the direction $\hat{\imath} + \hat{\jmath}$ at $P(1,1)$.

Ans : $3\sqrt{2}$

Example 15: Find the directional derivative of $f(x,y,z) = 2x^2 + 3y^2 + z^2$ at the point (2,1,3) in the direction of the vector $i - 2k$.

Ans :$-\frac{4}{\sqrt{5}}$

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